Gradient-Boosted Trees and Consensus-Risk

For Opioid and Polypharmacy Emergency Department Visits in Virginia's APCD — Final Dissertation Defense

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Integrative Life Sciences Doctoral Program (Personalized Medicine Track · Home Department)
School of Pharmacy (Department of Pharmacotherapy & Outcomes Science · Research Lab)
Virginia Commonwealth University

November 23, 2026

VCU School of Life Sciences and Sustainability | VCU School of Pharmacy

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VCU Integrative Life Sciences
Personalized Medicine Track · Genomics, Bioinformatics & Quantitative Systems Biology

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THE DISSERTATION ARCHWAY — **6 CHAPTERS · 5 CTS PAPERS**
Consensus-Risk Engine (Big Data → Bedside CDS)
Bridging `1.8 TB` Virginia APCD Claims, Symbolic SAT/SHAP XAI & Serverless `6 ms` PGx Triage

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Dissertation Defense Roadmap: The Clinical OODA Loop

FINAL DEFENSE · 6 CHAPTERS · 5 CTS MANUSCRIPTS

🎯 ANCHOR: FROM 4.2M POPULATION RECORDS (1.8 TB) TO 1 BEDSIDE PATIENT (6 MS)

1. OBSERVE — CHAPTER 1

PRISMA 2020 SQLR

`9,571` screened → `94` ML studies; `81%` PGx omission, `30%` leakage.

`n = 0 / 73 PROBAST`

➔

2. ORIENT — CHAPTER 2

Lakehouse, Audit & 5-Set FFA

`DuckDB` Spot-EC2 (`15.1x` speedup, `99%` cost drop) + Step 3b Purge (`348` codes).

`$23k/mo ➔ $6-$10/pass`

➔

3. DECIDE — CHAPTERS 3 & 4

Consensus GBD Holdouts

Ch. 3: `2.3x-4.4x` lift, `6.8-mo` DTW.

Ch. 4: Table S5 `60%` FFA pairs kept.

`OR = 0.25 (Q2 Z-Codes)`

➔

4. ACT — CHAPTERS 5 & 6

Serverless CDS & FDA SaMD

`56` ensembles in Lambda (`0` PHI); $|\Delta\hat{p}| < 0.06$ at `70%` sparsity; 38yo SNP case.

`6 ms · -22 pt Risk Drop`

Chapter	Clinical OODA Loop	Core Methodological & Empirical Milestones	Companion Manuscript Status
Ch. 1	1. Observe	Screened `9,571` records; 5 field gaps; `n = 0 / 73` **PROBAST models** unite PGx + XAI + temporal holdouts	**Under Review** · *J Pers Med* `jpm-4558734`
Ch. 2	2. Orient	`63` `Age Band × Year` DuckDB slices (`18.1M rec/min`); `348` post-index codes purged; 5-Set `Hitman` solver	**Under Consideration** · *J Big Data* `c26d4dae-…f18437`
Ch. 3★	3. Decide (Opioid)	Opioid ED (`N = 207,350`): 2019 holdout `AUROC 0.800-0.890` , Gate ablation `-61%` load (`112 → 44`), `6.8-mo` DTW	**Published** · *Clin Transl Sci* `10.1111/cts.70690`
Ch. 4★	3. Decide (Geriatric)	Geriatric Polypharmacy (`N = 647,002`): Table S5 `60%` FFA pair retention under `SHAP Jaccard = 0.00` ; Z-code `OR = 0.25`	**Published** · *Clin Transl Sci* `10.1111/cts.70718`
Ch. 5★	4. Act (System)	Serverless Dashboard Architecture + Live Interface (`6 ms` warm, `<$5/mo` , `0` PHI); `Imputation of Normality`	**Published** · *Clin Transl Sci* `10.1111/cts.70697`
Ch. 6	4. Act (Synthesis)	Dashboard Use Cases (`Tabs 1-5`) + Gene-Drug Network; 38yo female SNP walkthrough (`-22 pts`); FDA SaMD `21 CFR 820`	Dissertation Translational Synthesis

4 Core Defense Anchor Arguments: **(1) Empty Intersection:** `0/73` PROBAST models · **(2) Methodological Honesty:** Step 3b purges `348` post-index codes (`0.98` leaked → `0.800-0.890` honest) · **(3) Symbolic Logic Rescues Attribution:** Table S5 retains `60%` FFA pairs despite `SHAP Jaccard = 0.00` · **(4) Translational Completion:** `Imputation of Normality` ($|\Delta\hat{p}| < 0.06$ at `70%` sparsity) in `6 ms` Lambda.

Chapter 1: PRISMA 2020 SQLR & The "Empty Intersection" (n = 0 / 73)

CHAPTER 1 · JOURNAL OF PERSONALIZED MEDICINE (JPM-4558734) | PROSPERO CRD420261354089 · 9,571 RECORDS SCREENED · 94 ML STUDIES · 73 PROBAST MODELS

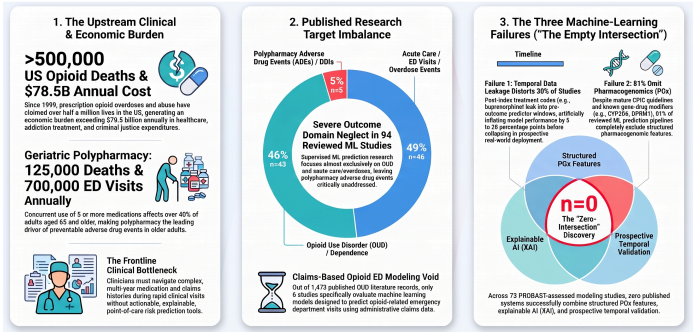
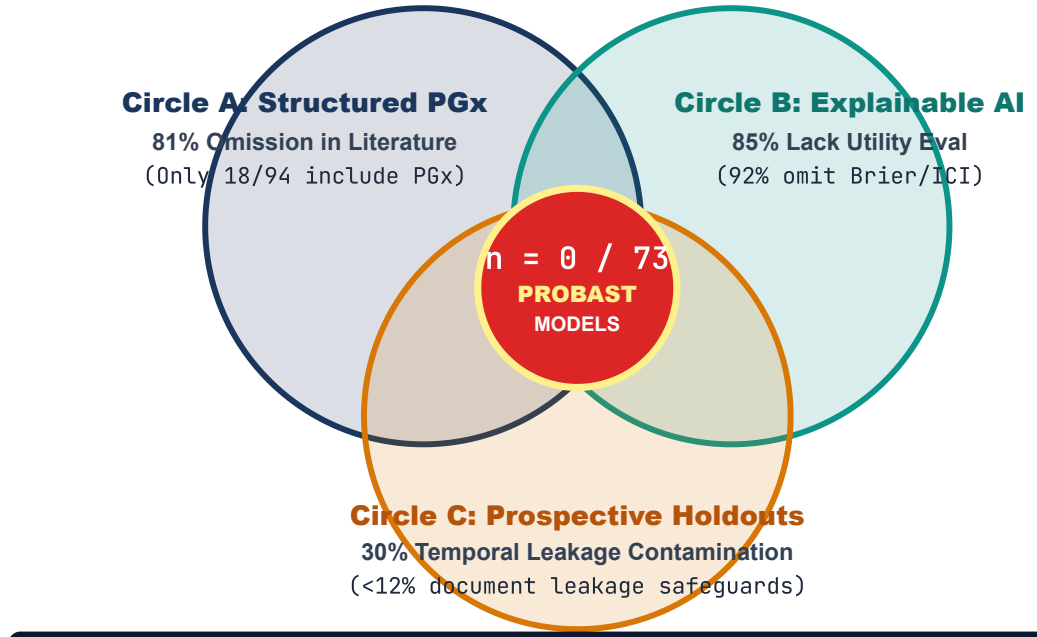


Figure 1.1 (Ch. 1): PRISMA 2020 SQLR (9,571 screened → 94 ML studies, 73 PROBAST models).

The Five Field-Wide Structural Gaps (2015-2026)

- **Gap 1 — PGx Omission (81%):** Only 19% (18/94) include any PGx feature despite CPIC Level A/B *CYP2D6 / OPRM1* guidelines.
- **Gap 2 — Temporal Leakage (~30%):** <12% (11/94) document leakage guards; ~30% use random CV with post-index contamination.
- **Gap 3 — Claims Neglect (2.4%):** Only 4 / 167 decision models use statewide administrative claims.
- **Gap 4 & 5 — XAI (85%) & Calibration (92%):** 85% lack clinical utility evaluation; 92% omit calibrated uncertainty (Brier / ICI); only 5% (n=5) study polypharmacy ADEs.

Infographic 1.1: 3-Way PROBAST Literature Venn Diagram ()



THIS DISSERTATION FILLS THE EMPTY INTERSECTION (PGx + Dual XAI + 2019 Holdo

Chapter 1: Dual-Cohort Funnel vs. "The Silent ADE Iceberg"

CHAPTER 1 · OPERATIONAL MAP FOR CHAPTERS 2–5 | VIRGINIA APCD (2016–2019) · 4.2M PATIENTS · 380M CLAIMS · 2.1B SERVICE LINES

LEFT FUNNEL · COHORT 1 (CH. 3 `opioid_ed`) — N = 207,350 (1:5 MATCHED)

▼ **The Acute OUD Surge Funnel (12-Month Lookback Window)**

- **Wide Young-Adult Net (`25-44` ★ = `107,388`):** Funneled through a **12-month longitudinal horizon** capturing chronic μ-opioid tolerance, dose escalation, and sedative co-prescribing (`gabapentinoids`, `benzodiazepines`).
- **Explicit Diagnostic Anchor (t_0):** ICD-10 `F11.20` & `T40.0X-T40.4X` scanned across all 10 positions (`26,710` cases / `180,640` controls).

RIGHT ICEBERG · COHORT 2 (CH. 4 `non_opioid_ed`) — N = 647,002 (17,287 GERIATRIC)

🧊 **The "Silent ADE" Iceberg (21–30 Day Proximal Window)**

- **Visible Tip Above Waterline (`<5%`):** Explicitly coded drug poisonings (`T36-T50`, `Y40-Y59`).
- **Submerged Mass Below Waterline (`>95%` "Silent ADEs"):** Downstream falls (`R29.6`), hip fractures (`S72`), syncope (`R55`), delirium (`F05`) & AKI (`N17`) linked via Milliman HCG `P51b/011/P33` + active Rx within **21–30 days pre-ED** (`0` opioid codes + `0-day` discharge guard).

Cohort & Chapter	Band 1 (`13-24` / `65-74`)	Band 2 (`25-44` / `75-84`)	Band 3 (`45-54` / `85-114`)	Band 4 (`55-64`)	Cohort Total (Cases / Controls)
`opioid_ed` **(Ch. 3 Funnel)**	`13-24`: 1,630 / 12,080 (13,710)	`25-44`: **12,753 / 94,635 (107,388) ★**	`45-54`: 5,984 / 37,655 (43,639)	`55-64`: 6,343 / 36,270 (42,613)	**26,710 Cases / 180,640 Controls (207,350)**
`non_opioid_ed` **(Ch. 4 Iceberg)**	`65-74`: **801 / 10,770 (11,571) ★**	`75-84`: 213 / 2,980 (3,193)	`85-114`: 168 / 2,355 (2,523)	*All 8 bands trained: N = 647,002*	**1,182 Cases / 16,105 Controls (17,287)**

Chapter 2 (Novelty #1A): Architectural Crossroads — Spark Tollbooth vs. DuckDB Express Lane

CHAPTER 2 · JOURNAL OF BIG DATA (SNAPP C26D4DAE-…F18437) | OVERTURNING DISTRIBUTED SPARK/EMR DOGMA ON 1.8 TB SKEWED CLAIMS

THE SPARK / AWS EMR TOLLBOOTH — MULTI-NODE SHUFFLE BOTTLENECK

Geriatric "Skew Tax" + Global Network Shuffles

65+ strata dominate claim volume → cross-node `toPandas` shuffle spikes & JVM-to-Python serialization friction.

`exit 137 OOM Crash (~$23,000/mo)`

PARTITION-FIRST DUCKDB EXPRESS LANE — 63 ISOLATED STORAGE TRACKS

Zero-Copy Arrow → Single 32-Core Fat-Spot EC2

63 `Age Band × Year` Parquet tracks feed 32 in-process `DuckDB` workers co-located with native `CatBoost` / `XGBoost` / `PySAT`.

`15.1x Throughput (18.1M rec/min) · $6-$10/pass (99% Drop)`

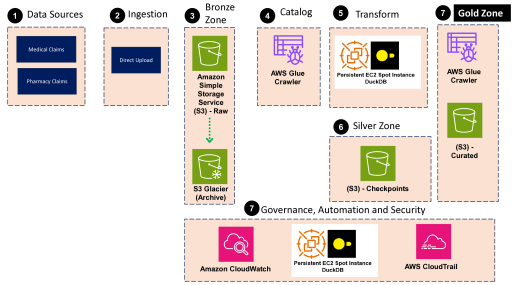


Figure 2.1 (Ch. 2): Partition-First Out-of-Core Medallion Pipeline (`ch02_npj.qmd` Fig. 3): `(1) Data Sources → (2) Ingestion → (3) Bronze Zone → (4) Glue Catalog → (5) Transform on Persistent EC2 Spot Instance DuckDB → (6) Silver Zone Checkpoints → (7) Gold Zone Curated` (`Age Band × Year`, `63` partitions).

1. Why Distributed Spark / AWS EMR Fails (`exit 137`)

- **Monolithic Saturation (`450 GB RAM`, `1.2M rec/min`):** Raw 2016–2019 files total `1.8 TB` (`6,929,576` unique patients).
- **Geriatric Skew & OOM Kills:** On multi-node EMR (`m5.8xlarge`, `c7g.16xlarge`), global shuffles across skewed `65-114` strata triggered **executor** `exit status 137` **OOM kills**.
- **JVM-to-Python Glue Gap:** Spark MLlib lacks parity with native `CatBoost`, `XGBoost-RF`, `Optuna` TPE, exact `TreeSHAP`, and `PySAT Hitman`.

2. Partition-First In-Process `DuckDB` + Fat-Spot EC2

- **Zero Cross-Node Shuffle (`63` `Age Band × Year` Slices):** Out-of-core `DuckDB` workers parallelize across 32 cores of **one Fat-Spot EC2 node (`x2iedn.8xlarge`)** with local NVMe spill. A skewed geriatric band never stalls a multi-node fleet!
- **Co-Located Native ML & Symbolic XAI:** Zero-copy Arrow buffers feed directly into native Python `CatBoost`, `XGBoost`, `Optuna`, `TreeSHAP`, and `PySAT Hitman`.

Chapter 2 (Novelty #1B): 3-Tier Medallion Pipeline & `<60s` Spot Interrupt Shield

CHAPTER 2 · PORTABLE CLOUD ARCHITECTURE (`ch02_npj.qmd`) | BRONZE → SILVER → GOLD CHECKPOINTING · `<60s` SPOT SHIELD · COHORT CONTRACTS

TIER 1 · BRONZE — RAW INGESTION
1.8 TB Claims + FAERS
Fixed-width medical/pharmacy extracts + local EHR augment point.

→

TIER 2 · SILVER — VALIDATED COHORT
5:1 Match & 0-Day Guard
Atomic `Age Band × Year` Parquet checkpoints (`63` slices).

→

TIER 3 · GOLD — ANALYSIS-READY
4-Bin Density Routing
Routes to `Low/Med/High/Extreme` ; trees see only ordinal `0-3` .

SPOT INTERRUPT SHIELD — ⏱ STOPWATCH
<60s Resume · 0% Loss
Idempotent S3 manifests: `0` corruption across `189/189` runs.

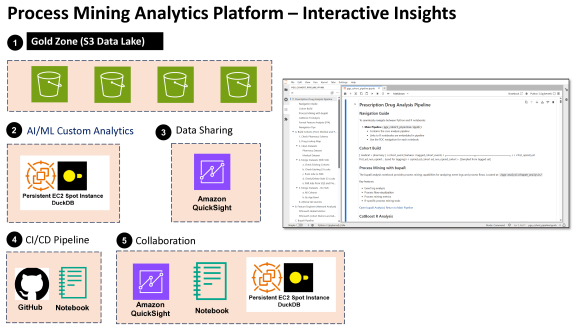


*Figure 2.2 (Ch. 2): **Gold Zone Analytics & Cohort Pipeline** (`pgx_cohort_pipeline.ipynb`): Persistent EC2 Spot Instance + `DuckDB` executing cohort builds, `bupaR` , `CatBoost` / `XGBoost` , and `FFA` with GitHub CI/CD.*

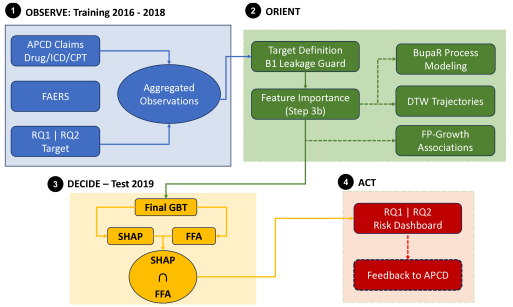


*Figure 2.3 (Ch. 2): **Clinical OODA Loop Pipeline Architecture** (`ch02_npj.qmd` Fig. 2): Governing `2016-2018` training, Step 3b B1 Leakage Guard, `SHAP ∩ FFA` Consensus Filter, and `2019` holdout feedback.*

S3 Medallion Lakehouse & `< 60s` Spot Checkpoint Recovery

- `Bronze` → `Silver` → `Gold` : Stores `1.8 TB` raw APCD + FAERS feeds, commits atomic `5:1` matched Parquet slices, and outputs partitioned feature matrices (`n_event_bin_ordinal` `0-3` , CPIC burden).
- **Spot Interruption Shield:** Resumes after EC2 Spot interruption in `< 60 seconds` **with** `0%` **data corruption**!

Portable "Cohort Contracts" for Low-Resource Health Systems

- **Separating Routing Volume from Tree Splits:** Raw `n_events` routes patients into `P25/P50/P95` density bins, while trees see only `n_event_bin_ordinal` (`0-3`) so visit volume cannot dominate splits.
- **Plug-and-Play Retargeting:** Safety-net hospitals can land local EHR/pharmacy Parquet extracts in `Bronze` and train local models via declarative **ICD-10 / Milliman HCG Cohort Contracts**.

Chapter 2 (Empirical Benchmarks): `15.1x` Speedup & `99%` Cloud Cost Collapse

CHAPTER 2 · EMPIRICAL SYSTEMS VALIDATION | 18.1M RECORDS/MIN · 0 PARTITION FAILURES · $6–$10 PER COMPLETED PASS

Table 2.1 (Ch. 2): DuckDB Worker Scaling vs. Monolithic & AWS EMR Baselines

Architecture / Configuration	Throughput (`rec/min`)	Speedup	Scaling Eff.	Failure / Recovery Profile
Monolithic Single-Node Scan	`1.2M / min`	`1.0x` (Ref)	100%	Saturated near **450 GB RAM**; monolithic stall
Multi-Node AWS EMR (Spark)	*Shuffle-bound*	—	Straggler-limited	**Executor OOM (** `exit 137` **)** on skewed 65+ bands
Partition-First DuckDB (8 Workers)	`9.4M / min`	`7.8x`	**97%**	0 failures; NVMe out-of-core spill active
Partition-First DuckDB (16 Workers)	`14.2M / min`	`11.8x`	**74%**	0 failures; S3 Parquet predicate pushdown
Partition-First DuckDB (32 Workers) ★	`18.1M / min`	`15.1x`	**47%** (I/O bound)	**0 partition failures across 3 full runs (** `63 × 3` **)**

Cloud Economics Collapse: AWS EMR vs. Fat-Spot EC2

Cost Metric	Multi-Node AWS EMR	DuckDB + Spot EC2 (`x2iedn.8xlarge`)
Peak Monthly Spend (Dec 2024)	`~$23,000 / mo` (`$5,200` EMR fee + `$17,800` EC2)	`~$277 / mo` (May 2025 full production month)
Failed Run Waste (May 2025)	`$1,340` (Terminated with shuffle `exit 137`)	`$0.00` (`<60s` checkpoint resume)
Cost Per Full Multi-Band Pass ★	`$400–$1,300+`	`$6.00 – $10.00 / pass` (~4 hrs @ `$1.50–$2.50/hr` Spot)

- **Proposition 1 Confirmed (** `Ch. 6` **):** Temporal validity and computational scalability are **not in conflict**—partitioning by `Age Band × Year` simultaneously yields a `15.1x` **speedup** and enforces **partition-isolated temporal cutoffs**!

Spot-Resilient Medallion Checkpointing (`< 60s` Recovery)

- **Idempotent S3 Manifests:** Each `Age Band × Year` slice commits atomically to `Bronze → Silver → Gold` on S3.
- **Spot Interruption Recovery:** Resumes from the exact interrupted partition in `< 60 seconds` with **zero artifact corruption** across all `189` evaluated partition invocations (`63 partitions × 3 full runs`).

Chapter 2 (Novelty #2): The Chronological Gatekeeper — Step 3b Leakage Audit

CHAPTER 2 · STEP 3B TEMPORAL AUDIT | 1,029 FEATURES → 348 PURGED (33.8%) → 682 SAFE

PRE-INDEX SAFE ZONE (t<t_0) — 66.2% WHITELISTED

682 Verified Antecedents Locked

`114` pure antecedents ($\text{pre} \geq 80\%$) + `567` mixed-timing chronic Rx & comorbidity codes (`safe_feature_filter.json`).

POST-INDEX BIOHAZARD ZONE ($t \geq t_0$) — 33.8% EJECTED

348 Post-Target Codes Purged ($\text{post} \geq 80\%$)

Ejects ED rescue & MAT induction: `SUBOXONE` (`6,082`, `100%`), `NARCAN` (`271`, `100%`), `CPT 80307` screens (`3,636`, `100%`).

DISCRIMINATION IMPACT GAUGE — ANCHOR #2

Deceptive 0.98 → Honest 0.800–0.890

Unaudited: `0.98 AUROC` (post-event leakage).
Purged Holdout: `0.800-0.890 AUROC` (`2.3x-4.4x` PR lift).

Empirical Post-Index Contamination (`348 / 1,029` Features Purged = `33.8%`)

Leakage Vector (`n` Purged)	Specific Feature / Code	Post-Index ($t \geq t_0$)	Post Ratio
1. Post-Diagnosis OUD Rescue & MAT Drugs (`164` drugs; `47.1%`)	`item_drug_SUBOXONE` ★	`6,082` events	`100.0%`
	`item_drug_BUPRENORPHINE_HCL/NALOXON`	`3,032` events	`100.0%`
	`item_drug_BUPRENORPHINE_HCL`	`2,562` events	`100.0%`
	`item_drug_ZUBSOLV` / `NARCAN` / `VIVITROL`	`400` / `271` / `106`	`100.0%`
	`CLONIDINE_HCL` & `BACLOFEN` (withdrawal)	Post-ED stabilization	$\geq 80\%$
2. Toxicology & Monitoring (`84` CPTs; `24.1%`)	`CPT 80307` (Presumptive drug screen) ★ `CPT 80305`, `80306`, `HCPCS G0480`	`3,636` events ED / MAT compliance	`100.0%` $\geq 80\%$
3. Follow-Up Billing (`100` ICDs; `28.7%`)	Subsequent OUD / rehab encounter codes	Downstream of t_0	$\geq 80\%$

Statewide Confirmation of Gal et al. (2023) & Holdout Impact

- **Statewide Claims Proof of VCU ED Workflow:**
  - Directly confirms **Ochalek, Gal et al. (*DADR*, 2023,** `PMC10102819` **)**: buprenorphine (`11,676` combined post-index events) and naloxone (`271`) occur `100%` **post-index ($t \geq t_0$)** as ED rescue/bridge induction!
- **Locked Safe Whitelist (`682` Features = `66.2%`):**
  - `114` **Pure Antecedents** ($\text{pre_ratio} \geq 0.80$) + `567` **Mixed-Timing** ($\text{pre} > 0, \text{post_ratio} < 0.80$) + `1` **Low-Freq Signal**.
- **Empirical Discrimination Correction (`Ch. 2-3`):**
  - **Unaudited Leaked Model:** `AUROC = 0.98` *(predicting yesterday's overdose from today's Suboxone & CPT 80307)*
  - **Leakage-Purged 2019 Holdout:** `AUROC = 0.800-0.890` (`0.76` pooled baseline; `PR Lift 2.3x-4.4x`)
- View Complete Step 3b Audit Breakdown ↗

Chapter 2 (Novelty #3A): The Dual-Funnel Consensus Filter & 5-Set PySAT Solver

CHAPTER 2 · STEPS 7-8 SYMBOLIC ARCHITECTURE | SOLVING THE O(2^N) PYSAT HITMAN BOTTLENECK & NON-CIRCULARITY PROBLEM

SIEVE 1 · STATISTICAL AI — **STEP 7 TREESHAP**

CatBoost ∩ Best XGBoost

Exact normalized `TreeSHAP` intersection ($\text{mean}\,|\text{SHAP}| \geq P_{75}$) across distinct gradient-boosting architectures.

∩

SIEVE 2 · SYMBOLIC LOGIC — **STEP 8 PYSAT HITMAN**

5-Set CNF Union → Minimal AXP

Bounds `>10,000` Horn clauses to `~300-500` rules (preserving `SHAP=0` gates), solving $O(2^n)$ stalls (`~10x` speedup).

➔

CATCH BASIN · TIER-1 CONSENSUS — **-61% REVIEW LOAD**

44 High-Conviction Clinical Rules

Ejects `252` **SHAP-only billing artifacts** & `33` **rare FFA overfits**; compresses pharmacist review `112 ➔ 44`.

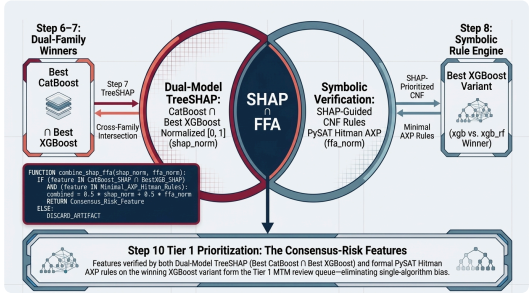


Figure 2.3 (Ch. 2): Implemented Consensus-Risk Pipeline (`Steps 6-8 & 10`): Step 7 Dual-Model `TreeSHAP` Gate (`CatBoost` ∩ `Best XGBoost`) → Step 8 DataFrame-Exploded CNF Compilation (`_explode_tree_to_dataframe`) → Bounded 5-Set Candidate Rule Union (`Set 1-5`) → `PySAT Hitman(solver="m22")` Minimal Hitting-Set AXP Extraction.

1. The Reference FFA Bottleneck (`Ignatiev et al., 2023`)

- **Unfiltered Path Explosion (`>10,000` Rules/Patient):** Feeding `100%` of satisfied root-to-leaf tree paths into NP-hard `PySAT Hitman` triggers `O(2^n)` runtime stalls on 500-tree APCD ensembles.
- **The Circularity Trap of Naive Top-SHAP Filtering:** Pre-filtering strictly by top $|\text{SHAP}|$ before `Hitman` makes `SHAP` ∩ `FFA` circular and discards `SHAP = 0` **logical threshold gates** required inside minimal AXPs (**Huang & Marques-Silva, 2024**).

2. Our Solution: Dual-Model Gate + DataFrame CNF + 5-Set Union

- **Step 7 Cross-Architecture `TreeSHAP` Gate:** Requires $\mathcal{F}_{\text{SHAP}} = \text{supp}(\text{SHAP}_{\text{CatBoost}}) \cap \text{supp}(\text{SHAP}_{\text{BestXGB}})$.
- **Step 8 DataFrame-Vectorized CNF Compilation:** Explodes root-to-leaf paths into tabular CNF intervals and forms a **bounded 5-Set Candidate Union (`~300-500` unique rules/patient, `~10x` solver speedup)** before invoking `Hitman(solver="m22")`.

Chapter 2 (Novelty #3B): The 5-Set Candidate Rule Filtering Framework (`Set 1-5`)

CHAPTER 2 · VCU INVENTION DISCLOSURE CLAIM 3 (`base_symbolic_explainer.py`) | ORIGINAL ENGINEERING DESIGN GROUNDED IN FOUNDATIONAL PEER-REVIEWED LITERATURE

Stage (`Step 8` Code)	Cap	Operational Specification & Bias / Completeness Safeguard	Foundational Literature Grounding
Set 1: First Matched `matched_rules[:100]`	`100`	Retains the first `100` matched rules in boosting tree-index order ($t = 1 \dots T_0$), capturing largest negative-gradient step sizes and primary main/interaction splits before residual fitting.	• **Friedman JH (2001)**, *Ann. Stat.* 29(5):1189–1232 (Gradient Boosting) • **Friedman & Popescu (2008)**, *Ann. Appl. Stat.* 2(3):916–954 (*RuleFit*)
Set 2: Random Sample `rng.choice(seed=42)`	`100`	Uniform random sample of `100` matched rules independent of SHAP score; guarantees deep-tree stochastic diversity and **prevents SHAP selection circularity**.	• **Breiman L (2001)**, *Mach. Learn.* 45:5–32 • **Lundberg SM et al. (2020)**, *Nat. Mach. Intell.* 2:56–67
Set 3: SHAP-Prioritized + Active Coverage ★ `top_300` ∪ `SHAP>0`	`300+`	Top `300` matched rules ($\geq 10\text{th}$ pct) ranked by instance $\lvert\text{SHAP}\rvert$, plus mandatory rule coverage for `100%` **of active** `SHAP > 0` **features**.	• **Lundberg SM, Erion G, Chen H, et al. (2020)**, *Nat. Mach. Intell.* 2(1):56–67 (Exact polynomial-time `TreeSHAP`)
Set 4: Zero-SHAP Fallback ★ `zero_shap[:100]`	`100`	Includes up to `100` matched rules whose features have $\phi_i(x) = 0$ (not covered by Sets 1–3), preserving **logical threshold gates required in minimal AXPs**.	• **Huang X & Marques-Silva J (2024)**, *Int. J. Approx. Reason.* 171:109112 (*Failings of Shapley Values*) • **Ignatiev A et al. (2023)**, `arXiv:2307.03380`
Set 5: Dataset-Frequent `top_100_freq_rules`	`100`	Pre-computes dataset-wide rule frequency (`compute_rule_frequencies()`) and retains top `100` matching rules, preserving reference FFA's population-frequency weighting.	• **Ignatiev A, Cooper MC, Marques-Silva J, et al. (2023)**, `arXiv:2307.03380` • **Izza Y, Ignatiev A, Marques-Silva J (2022)**, *JAIR* 75:261–321
Post-Solver Gate 2 ★ `Supp ⩾ 0.05, Conf ⩾ 0.70`	**AXP**	Filters minimal Class-1 AXP Horn rules extracted by `Hitman(solver="m22")` at $\text{Support} \geq 0.05$ (`5%`) and $\text{Confidence} \geq 0.70$, intersected with `mean |SHAP| ⩾ P75` .	• **Bénard C, Biau G, Da Veiga S, Scornet E (2021)**, *Electron. J. Stat.* 15:427–505 (**SIRUS** $p_0 \approx 0.05$ rule stability proof) • **Liu B et al. (1998)**, *KDD* (**CBA** support/confidence)

Dissertation-Wide Pruning Impact (Ch. 2): Intersecting `Gate 1 (SHAP ⩾ P75)` with `Gate 2 (5-Set FFA Support ⩾ 0.05, Confidence ⩾ 0.70)` retained **~49% of candidates** (`384-498` in `opioid_ed` ; `29-89` in geriatric `non_opioid_ed`), removing **252 high-frequency SHAP-only documentation artifacts** and **33 low-signal FFA-only rules** , and reducing holdout Brier variance by **0.008–0.013** .

Chapter 2 (Novelty #3C): Consensus Math, Intervention Rate (IR) & Brier Pruning

CHAPTER 2 · STEPS 7–8 & 10 MATHEMATICAL FORMULATION (ch02_npj.qmd) | REMOVING 252 SHAP-ONLY UTILIZATION ARTIFACTS & 33 RARE FFA OVERFITS

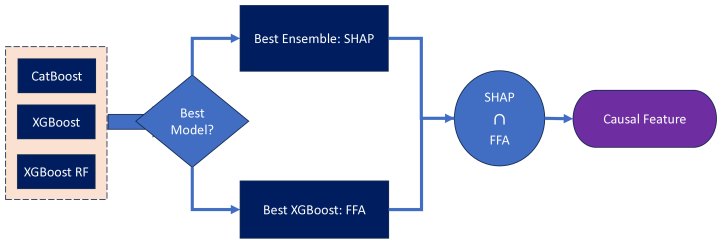


Figure 2.5 (Ch. 2): Dual-confirmation Consensus Filter ($\mathrm{SHAP} \geq P_{75} \cap \mathrm{FFA}_{0.05,0.70}$).

Step 8 Responsibility & Counterfactual Intervention Rate (IR)

- **Combined Synthesis (Step 10):**

$$\mathrm{CausalResp}(f) = \frac{\mathrm{RuleFreq}(f)}{\sum_k \mathrm{RuleFreq}(f_k)} \times \mathrm{SHAP}_{\mathrm{consensus}}(f)$$

- **Counterfactual Intervention Rate (IR > 0.10):** Measures how often setting feature j to its training median changes the patient's minimal Abductive Explanation (AXP):

$$\mathrm{IR}(j) = \frac{\left|\left\{i \in S_j : \mathrm{AXP}\left(x_i^{(j \to \mathrm{med})}\right) \neq \mathrm{AXP}(x_i)\right\}\right|}{|S_j|}$$

Empirical Pruning & 2019 Holdout Calibration Gain (Ch. 2)

Attribution Regime	Features Retained	Discrepant Artifacts Removed / Admitted	2019 Holdout Median Brier Range
Consensus Filter ★ SHAP ≥ P75 ∩ FFA (0.05, 0.70)	~49% **of candidates** (498 opioid / 89 geriatric)	Retains only dual-confirmed clinical & PGx drivers across tournament winners	0.008 - 0.051 (*Lowest variance*)
SHAP-Only Subset (Positive SHAP, Failed FFA)	+252 features	252 **E/M utilization & documentation codes** lacking structural tree-rule purity	0.014 - 0.068 ($\Delta = +0.006$–0.017)
FFA-Only Subset (Passed FFA, Below SHAP P75)	+33 features	33 **rare drug combinations (** <2% **prevalence)** with weak global attribution	0.011 - 0.074 ($\Delta = +0.003$–0.023)

- **Visualization-Only Process Mining Guard:** FP-Growth and bupaR run **exclusively on Consensus Filter codes** for pathway visualization—preventing the **15–20 percentage-point** AUC-PR **target-leakage inflation** caused by feeding unconstrained association rules into tree predictors!

Chapter 3: Opioid ED 2019 Prospective Holdout & Metric Gauge Array

CHAPTER 3 · PUBLISHED IN CLINICAL AND TRANSLATIONAL SCIENCE (DOI: 10.1111/CTS.70690) | 26,710 CASES / 180,640 CONTROLS · UNTOUCHED 2019 HOLDOUT (PR LIFT 2.3X–4.4X)

⏱ GAUGE 1 · AGES 13–24 — AUROC 0.809
2.3× PR Lift over Prevalence
`PR-AUC 0.375` · `Brier 0.008` · `384` rules

⏱ GAUGE 2 · AGES 25–44 ★ — PRIMARY (N=4,942)
3.3× PR Lift (AUROC 0.800)
`PR-AUC 0.557` · `Brier 0.013` · `498` rules

⏱ GAUGE 3 · AGES 45–64 — AUROC 0.842–0.864
2.9× – 3.4× PR Lift
`PR-AUC 0.478–0.559` · `498` rules

📐 GAUGE 4 · AGES 65+ & CALIB. — AUROC 0.824–0.890
2.9× – 4.4× Lift · Actionable p̂
`Brier 0.003–0.004` · `846` FFA features

Table 3.1 (Ch. 3): `opioid_ed` 2019 Holdout & Consensus Drivers

Age Band	Selected Winner	AUROC	PR-AUC	PR Lift	Recall	Brier	ICI	Consensus Features
`13–24`	Ensemble	`0.809`	`0.375`	`2.3x`	`0.524`	`0.008`	`0.164`	`384`
`25–44` (Primary) ★	**Ensemble**	`0.800`	`0.557`	`3.3x`	`0.647`	`0.013`	`0.108`	`498` (`44` low)
`45–54`	Ensemble	`0.842`	`0.559`	`3.4x`	`0.687`	`0.007`	`0.154`	`498`
`55–64`	Ensemble	`0.864`	`0.478`	`2.9x`	`0.778`	`0.051`	`0.138`	`498`
`65–74`	Ensemble	`0.824`	`0.479`	`2.9x`	`0.738`	`0.003`	`0.147`	`385` (FFA)
`75–84` / `85–114` [a]	Ensemble	`0.886` / `0.890`	`0.629` / `0.741`	`3.8x` / `4.4x`	`0.830` / `0.689`	`0.004` / `0.003`	`0.206` / `0.209`	`292` / `169` (FFA)

[a] Age-band aggregate holdout. Primary `25–44` low-density holdout evaluates $n = 4,942$ untouched 2019 cases (`0.98` leaked → `0.800–0.890` honest).

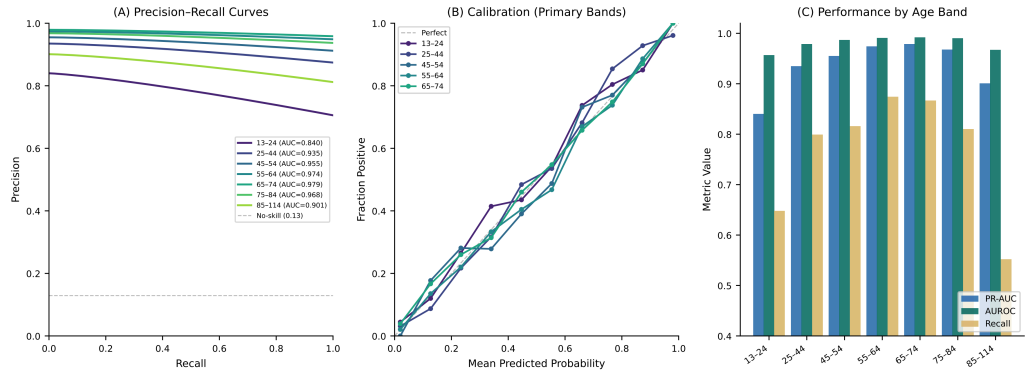


Figure 3.1 (Ch. 3): 2019 Prospective Holdout ROC, Precision-Recall & 45° Calibration Curves (`2.3x–4.4x` PR lift; `Brier 0.003–0.051`).

Chapter 3 Sensitivity Analysis: Two-Gate Mirror Ablation (CTS-2026-0196)

CHAPTER 3 · PEER-REVIEW SENSITIVITY ANALYSIS (DOI: 10.1111/CTS.70690, REVIEWER #2 MAJOR COMMENT 6)

TESTING SIGNAL RETENTION, BOUNDARY NOISE FILTERING & REVIEW-LOAD COMPRESSION ($n = 181$ FEATURES)

Gate Regime (25-44 low , $n = 181$)	Features Passing	Top-20 SHAP Retention	Boundary / Discrepant Features	Operational & Clinical Impact
1. Consensus-Risk Intersection ★ SHAP ⩾ P75 ∩ FFA (0.05, 0.70)	44 / 181 (24.3%)	20 / 20 (100%) (pgx_num_drugs , gabapentin , pgx_num_cpic_drugs)	0 **top-20 drivers lost**	**Optimal Operating Point:** 100% core signal in a compact, auditable 44-feature Tier-1 pharmacist list.
2. Gate 1 Alone (SHAP-Only) SHAP ⩾ P75 (No FFA Filter)	46 / 181 (25.4%)	20 / 20 (100%)	2 **features pass SHAP but fail FFA:** • item_drug_BUNAVAIL (0.0024) • item_drug_BUTRANS (0.0019)	**Boundary Noise Filter:** FFA rejects 2 sparse buprenorphine/OUD correlates at the P75 tail lacking support ($< 5\%$) or purity ($< 70\%$).
3. Gate 2 Alone (FFA-Only) FFA (0.05, 0.70) (No SHAP Filter)	112 / 181 (61.9%)	20 / 20 (100%)	68 **features pass FFA but fall below** SHAP P75	**Review-Volume Compression (61% Reduction):** Without Gate 1, clinician faces $2.5\times$ **more rules (112 vs. 44)** of routine background context clauses.

Three Sensitivity Questions Answered Empirically

1. **Does FFA (0.05, 0.70) clip true clinical drivers? No** — 20 / 20 (100%) of the top-20 SHAP drivers pass the symbolic FFA gate.
2. **What does FFA reject at the SHAP P75 boundary?** Exactly 2 **residual buprenorphine correlates (BUNAVAIL , BUTRANS)** sitting at the 75th-percentile tail ($\text{mean } |\text{SHAP}| = 0.0019$–$0.0024$).
3. **Why not use FFA alone?** Intersecting with SHAP ⩾ P75 compresses the pharmacist review set by 61% (112 → 44 features).

Cross-Cohort Scaling with Fixed Operational Thresholds

- Holding mean |SHAP| ⩾ P75 ∩ FFA (0.05, 0.70) fixed across all cohorts scales naturally with lookback dimensionality (consensus_risk_feature_counts_by_cohort_age_band.csv):
  - opioid_ed **(12-mo lookback):** 384 (13-24), 498 (25-44 , 45-54 , 55-64) across all density bins (44 in focal 25-44 low bin).
  - non_opioid_ed **(21-day proximal):** 89 (65-74), 33 (75-84), 29 (85-114).

Chapter 3 (Part 3): "HOW to Intervene" — Consensus Drivers & Claims-Recovered PGx

CHAPTER 3 · PUBLISHED IN CLINICAL AND TRANSLATIONAL SCIENCE (DOI: 10.1111/CTS.70690)

SHAP ∩ FFA SPECIFIES HOW TO INTERVENE · PROVING PROPOSITION 2 (CLAIMS-RECOVERED PGX)

DRIVER #1 · PGX BURDEN — MEAN |SHAP| = 0.83

`pgx_num_drugs` **(#1 Overall)**

Total concurrent pharmacogenomically active substrate load (`100%` FFA rule support).

+

DRIVER #2 & #3 · CNS SYNERGY — MEAN |SHAP| = 0.27

Gabapentin + Benzodiazepines

`gabapentin` (`0.27`), `clonazepam` (`0.08`), `alprazolam` (`0.05`) potentiate μ-opioid respiratory depression.

+

DRIVER #4 · PROPOSITION 2 PROOF — MEAN |SHAP| = 0.07

`pgx_num_cpic_drugs` **(No Genotype!)**

Proves CPIC Level A/B risk signal is recoverable from claims (`74.2%-85.9%` cohort coverage).

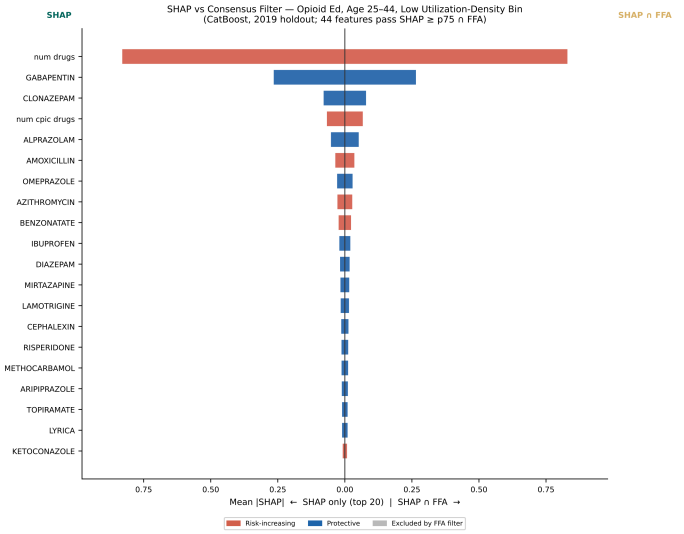


Figure 3.2 (Ch. 3): Top Consensus-Risk features in primary `25-44` band (`n = 4,942` untouched 2019 holdout cases).

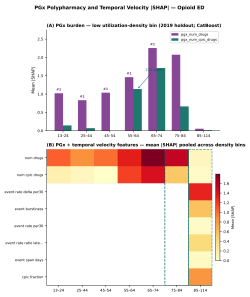


Figure 3.3 (Ch. 3): Monotonic non-linear `TreeSHAP` risk escalation as concurrent `pgx_num_drugs` and `pgx_num_cpic_drugs` exceed ≥ 4 substrates.

Rank & Feature (`25-44`)	Mean $\lvert SHAP \rvert$	FFA Support	Pharmacological Mechanism
#1 `pgx_num_drugs` ★	`0.83`	`100%`	Cumulative CYP2D6 / CYP3A4 / OPRM1 substrate saturation
#2 `item_drug_GABAPENTIN` ★	`0.27`	`100%`	$\alpha_2\delta$ calcium-channel synergy; abolishes μ-opioid respiratory ceiling
#3 `CLONAZEPAM / ALPRAZOLAM`	`0.08 / 0.05`	`94%`	$GABA_A$ positive allosteric CNS & brainstem sedation
#4 `pgx_num_cpic_drugs` ★	`0.07`	`92%`	CPIC Level A/B actionable gene-drug burden (Proposition 2)

Chapter 3 (Part 4): "WHEN to Intervene" — DTW Archetypes & `6.8-Month` Window

CHAPTER 3 · PUBLISHED IN CLINICAL AND TRANSLATIONAL SCIENCE (DOI: 10.1111/CTS.70690)

DYNAMIC TIME WARPING (N = 59,813 TRAJECTORIES) · 6.8-MONTH (~27-WEEK) ACTIONABLE WINDOW

ARCHETYPE 1 · 96.4% OF COHORT — EPISODIC CARE

Low-Utilization Baseline

Intermittent outpatient fills with low cumulative gabapentinoid/opioid velocity.

➔

ARCHETYPE 2 · 1.7% (HIGH-VELOCITY) ★ — MEDIAN 21 CLAIMS

Gabapentin-Enriched Escalation

Rapid gabapentinoid + opioid co-prescribing acceleration prior to index ED overdose.

➔

ACTIONABLE SURVEILLANCE HORIZON — ~27 WEEKS PRE-ED

6.8-Month Lead Time (10.9% → 18.4%)

ED risk surges from `10.9%` (DTW Q1) to `18.4%` (DTW Q3) at Month -6.8 before presentation!

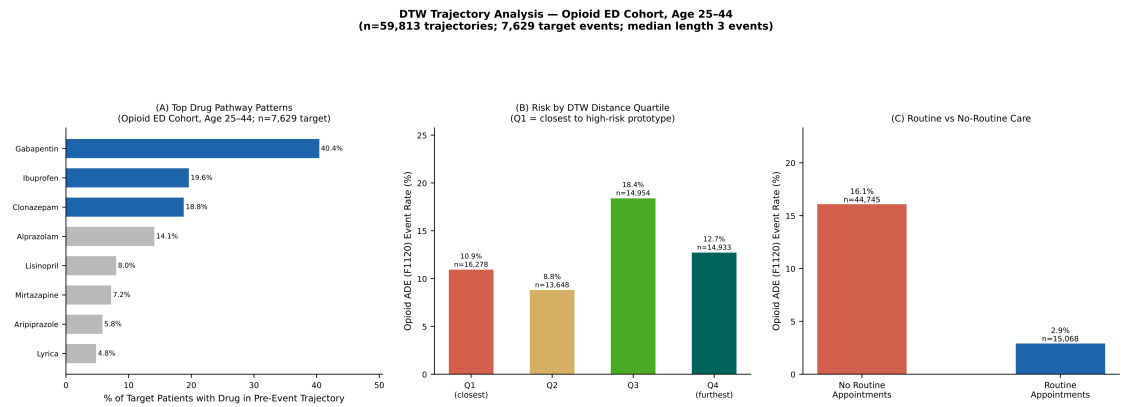


Figure 3.4 (Ch. 3): Dynamic Time Warping (DTW) patient trajectory archetypes (`n = 59,813` in `25-44`) and `6.8-month` pre-index window.

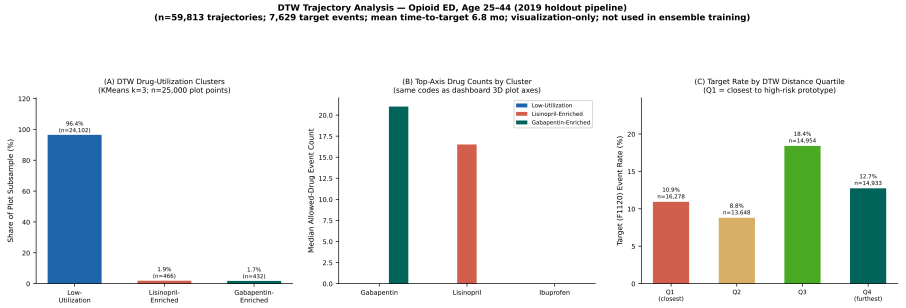


Figure 3.5 (Ch. 3): Pre-index prescription density trajectories showing inflection at Month -6.8 (~27 weeks prior to t_0).

Why `DTW` Complements `SHAP ∩ FFA` (`78%` Holdout Concordance)

- `SHAP ∩ FFA` = *HOW* to Intervene: Identifies which specific drugs (`gabapentin`, `clonazepam`, `pgx_num_cpic_drugs`) drive patient risk.
- `DTW` (Sakoe-Chiba $w = 20\%$) = *WHEN* to Act: Separates Rapid-Onset (`4.2 mo`, `74%` ED within 6 mo → immediate 30-day PDMP review) from Chronic-Escalation (`22.1 mo`, `81%` after 12 mo → physical therapy & taper window).

Chapter 3 (Part 5): Translating the "How & When" Opioid ED Triage Workflow

CHAPTER 3 · CLINICAL & HEALTH-SYSTEM TRANSLATION (DOI: 10.1111/CTS.70690) | CONNECTING STATEWIDE CLAIMS SURVEILLANCE TO OUTPATIENT PHARMACIST & PDMP ACTION

STEP 1 · DTW WINDOW DETECTION (WHEN) — MONTH −6.8 (~27 WEEKS)
Proactive PDMP & Payer Surveillance Trigger
Flags patients entering the `Gabapentin-Enriched` (`1.7%`) or `Rapid-Onset` (`4.2 mo`) trajectory **27 weeks before acute ED overdose**.

→

STEP 2 · PATHWAY REDIRECTION (WHERE) — −58% ESCALATION
Physical Therapy Prior to Gabapentinoid Initiation
`bupaR` care pathways prove patients receiving **physical therapy before gabapentin** had a `58%` **lower transition rate** to opioid escalation!

→

STEP 3 · CONSENSUS & PGX ACTION (HOW) — 44 TIER-1 RULES
Targeted Deprescribing & CPIC Genotyping Referral
Reviews compact `44-rule` Consensus list (`-61%` alert burden), tapers `gabapentin` + `benzodiazepines`, and triggers *CYP2D6 / OPRM1* PGx testing.

Clinical Question	Methodological Module	Empirical Chapter 3 Finding	Actionable Bedside / Payer Protocol
1. Who is at immediate risk? ★	4-Tier Density Ensemble (`CatBoost + XGB + XGB-RF`)	`AUROC 0.800-0.890` · `PR Lift 2.3x-4.4x` · `Brier 0.003-0.051`	Stratifies patients into calibrated Low (`<20%`), Moderate (`20-60%`), or High (`>60%`) risk bands
2. How do we prevent alert fatigue?	Two-Gate Consensus (`SHAP ⩾ P75` ∩ `FFA 0.05, 0.70`)	`61%` **compression (** `112 → 44` **rules)**; `20/20` (`100%`) core drivers kept	Delivers a concise, noise-free 44-item pharmacist review checklist
3. Which drugs drive acute toxicity? ★	Consensus Attribution + Step 5 CPIC Proxies	`#1 pgx_num_drugs` (`0.83`), `#2 gabapentin` (`0.27`), `#3 benzos` , `#4 pgx_num_cpic_drugs`	Prioritizes gabapentinoid/benzo taper & flags patients with ≥ 4 CPIC substrates for genotyping
4. When & where should we intervene? ★	`DTW` (`Sakoe-Chiba`) + `bupaR` Sankey Sequences	`6.8-month` **window (** `10.9% → 18.4%` **);** PT before gabapentin = `-58%` **escalation**	Routes musculoskeletal pain to PT *before* gabapentin and initiates taper 6.8 months pre-ED

Chapter 4 (Part 1): Geriatric Polypharmacy 2019 Holdout Performance

CHAPTER 4 · PUBLISHED IN CLINICAL AND TRANSLATIONAL SCIENCE (DOI: 10.1111/CTS.70718)

1,182 GERIATRIC CASES / 16,105 CONTROLS (647,002 ALL 8 BANDS) · HONEST LEAKAGE-CORRECTED HOLDOUT

⏱ GAUGE 1 · AGES 65–74 ★ FOCAL (N=801)
3.9× PR Lift (AUROC 0.877)
`AUPRC 0.301` · `Brier 0.008` · `89` rules

⏱ GAUGE 2 · AGES 75–84 AUROC 0.858
4.2× PR Lift (AUPRC 0.335)
`213 / 2,980` · `Brier 0.007` · `33` rules

⏱ GAUGE 3 · AGES 85–114 AUROC 0.686
1.6× PR Lift · Frailty Ceiling
`168 / 2,355` · `Brier 0.007` · `29` rules

📐 GAUGE 4 · AGES 13–64 PRE-GERIATRIC (N=629K)
2.6× – 3.8× Lift (AUPRC 0.268)
`79–153` FFA features · `85,311` rules

Table 4.1 (Ch. 4): Geriatric `non_opioid_ed` Holdout & Consensus Drivers

Cohort Band	Cases / Controls	Corrected AUROC	Corrected AUPRC (Lift)	Brier / ICI	Consensus Features
`65–74` **(Focal) ★**	`801 / 10,770`	`0.877`	`0.301` (`3.9x`)	`0.008 / 0.071`	`89`
`75–84`	`213 / 2,980`	`0.858`	`0.335` (`4.2x`)	`0.007 / 0.290`	`33`
`85–114`	`168 / 2,355`	`0.686`	`0.101` (`1.6x`)	`0.007 / 0.212`	`29`
`13–64` (Pre-Geriatric)	`629,715` total	`0.764–0.852`	`0.130–0.268` (`2.6x–3.8x`)	`0.006–0.019`	`79–153` (FFA)

- **Honest Leakage-Corrected Holdout (`N = 647,002` All 8 Bands):** Strict 2019 holdout after Step 3b purge & 4-tier density stratification (`PR lift 1.6x–4.2x` ; `85–114` reflects competing frailty hazards).

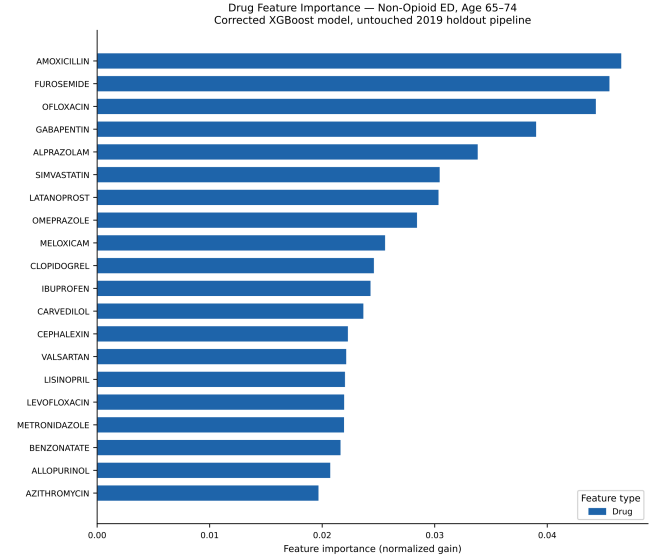


Figure 4.1 (Ch. 4): Geriatric Consensus-Risk attribution (`65–114`).

Chapter 4 (Part 2): Utilization-Free Stress Test (Table S5) — FFA vs. SHAP

CHAPTER 4 · PEER-REVIEW SENSITIVITY ANALYSIS (DOI: 10.1111/CTS.70718, AE COMMENT 2 & TABLE S5)

STRIPPING 17 UTILIZATION COVARIATES & 4-TIER DENSITY STRATIFICATION (3_model_sensitivity.ipynb)

Age Band	Primary Stratified Holdout AUPRC (Lift)	Utilization-Free Unstratified AUPRC (Lift)	ΔAUPRC	Top-25 Drug SHAP Jaccard (Gate 1 Stability)	Published FFA Synergistic Pairs Retained with $\mathrm{IE} > 0$ (Gate 2 Stability)
13-24 / 25-44	0.130 (2.6x) / 0.264 (3.5x)	0.068 / 0.032	-0.062 / -0.232	– / 0.09	Exploratory pre-geriatric / 2 / 5 (40%)
45-54	0.251 (3.4x)	0.068 (2.57x)	-0.183	0.09 (Low)	3 / 5 (60% retained: levofloxacin-lorazepam , carvedilol-levofloxacin , gabapentin-levofloxacin)
55-64	0.155 (1.9x)	0.072 (3.60x)	-0.083	0.00 (Complete Collapse)	3 / 5 (60% retained: levofloxacin-lorazepam , carvedilol-levofloxacin , gabapentin-levofloxacin)
65-74 (Focal) ★	0.301 (3.9x)	0.055 (1.95x)	-0.246	0.00 (Complete Collapse)	3 / 5 (60% retained: levofloxacin-lorazepam , carvedilol-levofloxacin , gabapentin-levofloxacin)
75-84	0.335 (4.2x)	0.163 (5.03x)	-0.172	0.09 (Near-zero)	3 / 5 (60% retained) (1/5 at strict CIs; 3/5` positive synergy)
85-114	0.101 (1.6x)	0.175 (5.42x)	+0.074	0.14 (Low)	3 / 5 (60% retained: levofloxacin-lorazepam , carvedilol-levofloxacin , gabapentin-levofloxacin)

🏚 CRUMBLING PILLAR · UNIVARIATE SHAP (GATE 1) — STATISTICAL FRAGILITY

Top-25 Drug SHAP Jaccard = 0.00 (Complete Collapse)

Stripping the **17 healthcare-utilization covariates** & 4-tier density stratification drops focal 65-74 AUPRC from 0.301 to 0.055 ($\Delta = -0.246$) and causes univariate additive SHAP drug rankings to completely reshuffle (Jaccard = 0.00 in 55-64 & 65-74).

⚔

🏗 REINFORCED STEEL BEAM · SYMBOLIC FFA (GATE 2) — COMBINATORIAL RESILIENCE

60% (3 / 5) Synergistic Drug Pairs Retained (IE > 0)

Despite SHAP Jaccard = 0.00 , 60% (3 / 5) **of published** FFA **pairs** (levofloxacin-lorazepam , carvedilol-levofloxacin , gabapentin-levofloxacin) survive across 45-74 & 85+ —proving formal logic isolates true pharmacotherapy co-exposure independent of billing density!

Chapter 4 (Part 3): 115 Synergistic Pairs, 312 Triplets & "Triple Whammy"

CHAPTER 4 · PUBLISHED IN CLINICAL AND TRANSLATIONAL SCIENCE (DOI: 10.1111/CTS.70718)

115 SYNERGISTIC PAIRS · 312 / 5,021 TRIPLETS ($\alpha_{\text{Bonf}} = 4.6 \times 10^{-5}$) · STOPP CRITERION D4

PAIR #1 & #2 · FLUOROQUINOLONE + CNS — IE = 16.3 & 11.9

Levofloxacin + Lorazepam / Acetaminophen

Fluoroquinolone GABA_A antagonism + benzodiazepine sedation triggers acute geriatric delirium & falls (`R29.6` , `S72`).

+

PAIR #3 · CARDIORENAL QT / HYPOTENSION — IE = 10.5

Carvedilol + Levofloxacin

Non-selective β/α_1-blockade + fluoroquinolone orthostatic syncope (`R55`) & arrhythmia cascade.

+

TOP 85–114 TRIPLET · STOPP CRITERION D4 ★ — 12.3% OF OLDEST-OLD CASES

"Triple Whammy": Furosemide + HCTZ + Lisinopril

Dual loop/thiazide volume depletion + ACEi efferent arteriolar dilation → acute kidney injury (`N17`) & hyperkalemia.

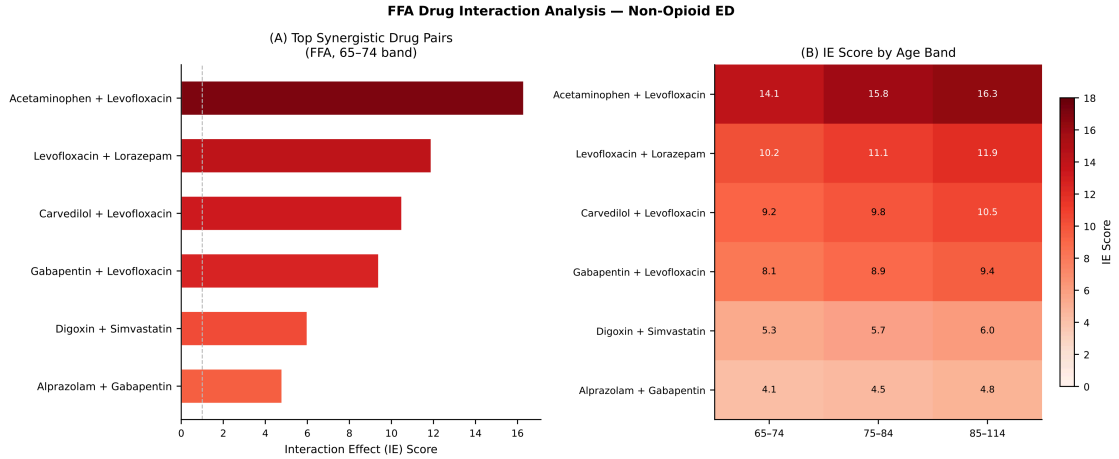


Figure 4.2 (Ch. 4): Formal Feature Attribution (FFA) Synergistic Drug Pairs (`115` pairs with positive Interaction Effect $\text{IE} > 0$).

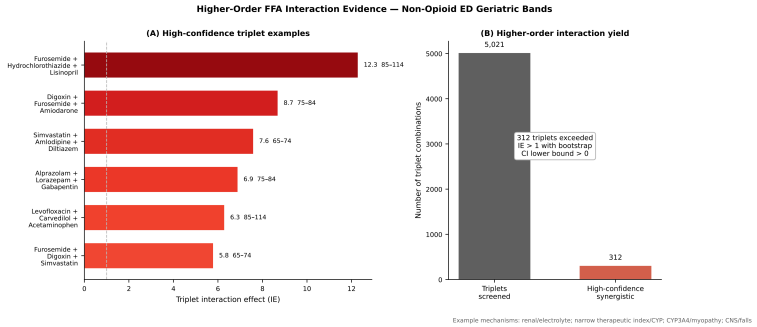


Figure 4.3 (Ch. 4): `312 / 5,021` (`6.2%`) statistically significant 3-drug triplets surviving Bonferroni correction ($\alpha_{\text{Bonf}} = 4.6 \times 10^{-5}$).

Why Combinatorial Horn Clauses Beat Pairwise DDI Tables

- Standard EHR drug-alert databases evaluate **2-drug pairs in isolation**, generating `>90%` overridden alerts.
- Our `PySAT Hitman` minimal hitting sets evaluate **higher-order 3-drug conjunctions** ($d_1 \wedge d_2 \wedge d_3 \Rightarrow \text{ED}$), recovering the **STOPP D4 "Triple Whammy"** (`furosemide + HCTZ + lisinopril`) in `12.3%` **of** `85-114` **cases** without hand-coded rules!

Chapter 4 (Part 4): Counterfactual Intervention Rate (`IR`) vs. Static Beers Lists

CHAPTER 4 · PUBLISHED IN CLINICAL AND TRANSLATIONAL SCIENCE (DOI: 10.1111/CTS.70718) | PATIENT-SPECIFIC DEPRESCRIBING PRIORITY (IR) MOVES BEYOND STATIC AGS BEERS CRITERIA

1. SYMBOLIC PERTURBATION EQUATION — STEP 8 & 10

$$\mathrm{IR}(j) = \Pr\left[\mathrm{AXP}(x_i^{(j\to\mathrm{med})}) \neq \mathrm{AXP}(x_i)\right]$$

Measures how frequently discontinuing drug j breaks the patient's minimal Abductive Explanation (AXP) rule.

→

2. TOP DEPRESCRIBING TARGETS (65–114) ★ — $\rho = 0.53$–0.68

#1 Simvastatin · #2 Furosemide · #3 Alprazolam

Followed by `levofloxacin` (`26.7%` pair IR), `lorazepam` , `carvedilol` , and `gabapentin` .

→

3. CLINICAL ADVANTAGE OVER STATIC BEERS — CONTEXT-AWARE

Ranks by Counterfactual Risk Drop ($\Delta\hat{p}$)

Distinguishes benign maintenance therapy from active multi-drug synergistic cascades (`IR > 0.10`).

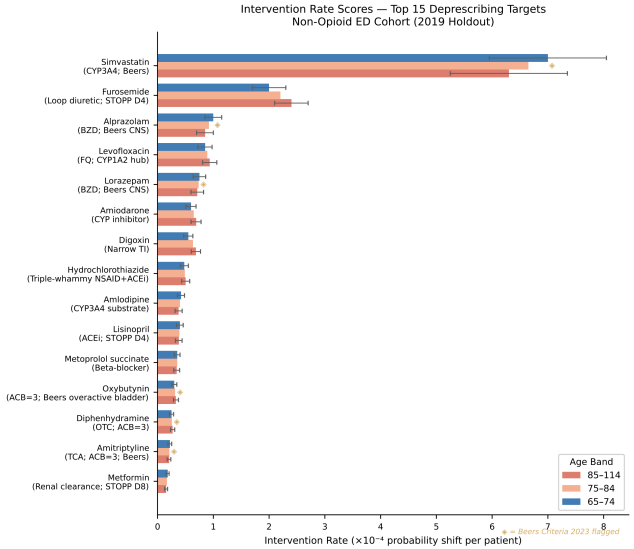


Figure 4.4 (Ch. 4): Counterfactual Intervention Rate (`IR`) rankings across geriatric age bands (`65–74` , `75–84` , `85–114`).

Deprescribing Target	Primary Mechanism in Multi-Drug AXP Rules	Pair / Single `IR`	Static AGS Beers (2023) Status
1. `Simvastatin` ★	*CYP3A4 / SLCO1B1* myopathy & fall cascade	**Rank #1**	Not flagged unless high-dose interaction
2. `Furosemide` ★	Loop diuretic dehydration, syncope & Triple Whammy AKI	**Rank #2**	Caution only (SIADH / volume depletion)
3. `Alprazolam / Lorazepam` ★	GABA_A sedation + fluoroquinolone synergy (`IR = 26.7%`)	**Rank #3 (`26.7%`)**	**Beers Potentially Inappropriate (Avoid)**
4. `Levofloxacin + Carvedilol`	Acute orthostatic hypotension, QT prolongation & delirium	`IR = 21.4%`	Missed by static single-drug Beers lists

Direct Translation into Dashboard Tab 3 (`IR > 0.10` Alerts)

- Whenever a clinician selects a medication on **Tab 3** belonging to an FFA pair/triplet with `IR > 0.10` , the dashboard fires an automated **Synergy Alert** quantifying the modeled risk reduction ($\Delta\hat{p}$) from deprescribing or substituting that agent.

Chapter 4 (Part 5): The "U-Shaped Shield" — Z-Code Protection (`OR = 0.25`)

CHAPTER 4 · PUBLISHED IN CLINICAL AND TRANSLATIONAL SCIENCE (DOI: 10.1111/CTS.70718)

NOVEL DISCOVERY #4: PREVENTIVE Z-CODE MONITORING SEPARATES MANAGED VS. UNMANAGED POLYPHARMACY

LEFT PEAK · Q1 (ZERO Z-CODES) — UNMANAGED CARE

10.7% Risk (OR = 1.00 Reference)

Unmonitored polypharmacy lacking documented preventive medication review or therapeutic drug monitoring.

SWEET-SPOT TROUGH · Q2 (1–12% Z-CODES) ★ — -75% RISK DROP

3.0% Risk (OR = 0.25, 95% CI: 0.18–0.34)

Active geriatric medication management shield—proves coordinated monitoring mitigates high drug counts!

RIGHT PEAK · Q4 (>20% Z-CODES) — FRAGMENTED CARE

10.0% Risk (OR = 0.94, Unmonitored Baseline)

Severely fragmented, multi-provider reactive care reverts to baseline ADE emergency visit risk.

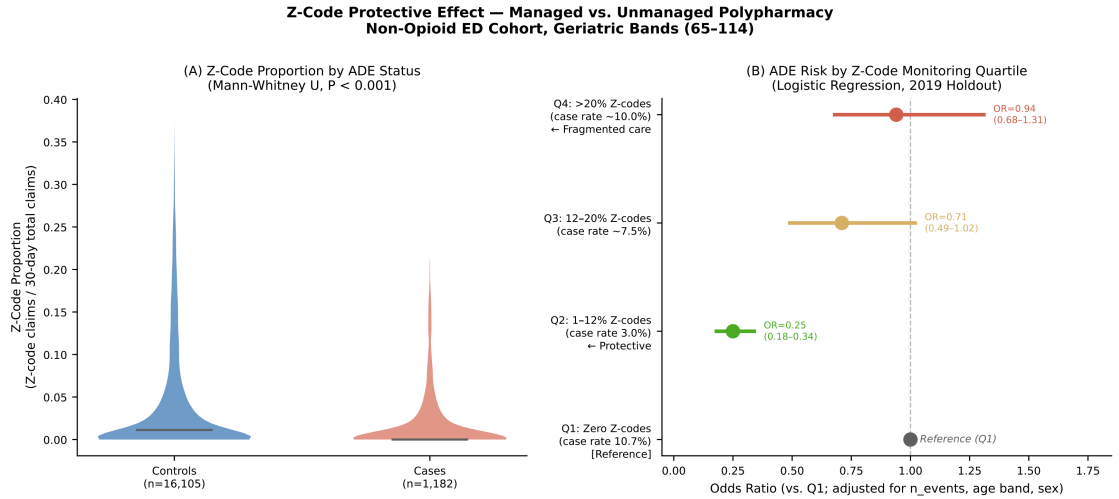


Figure 4.5 (Ch. 4): Novel U-Shaped Preventive Monitoring (`Z-Code`) Effect across geriatric strata (`OR = 0.25` in Q2).

Preventive Z-Code Quartile	Z-Code Share	Observed ADE ED Risk	Adjusted Odds Ratio (`95% CI`)
Q1: Zero Monitoring	`0%` (`0` Z-codes)	`10.7%`	`1.00` (Reference)
Q2: Coordinated Oversight ★	`1–12%` **(`1–2` Z-codes)**	`3.0%`	`0.25` **(`0.18–0.34`)** · `-75%`
Q3: Moderate-High	`12–20%` (`3–5` Z-codes)	`6.4%`	`0.58` (`0.44–0.76`)
Q4: Fragmented Care	`>20%` (`6+` Z-codes)	`10.0%`	`0.94` (`0.73–1.21`)

Paradigm Shift for Geriatric Deprescribing Policy

- *"Deprescribing based purely on raw pill count is flawed; proactive coding and coordinated monitoring (`Q2` , `OR = 0.25`) empirically separates safe **managed polypharmacy** from high-risk **unmanaged polypharmacy**."*

Chapter 5 (Part 1): Serverless Dashboard Architecture & Live Clinical Interface

CHAPTER 5 · PUBLISHED IN CLINICAL AND TRANSLATIONAL SCIENCE (DOI: 10.1111/CTS.70697)

SERVERLESS ARCHITECTURE (S3 · CLOUDFRONT · API GATEWAY · LAMBDA) + LIVE DASHBOARD UI

1. SERVERLESS ARCHITECTURE DIAGRAM — 0 PHI STORED · <$5/MO

S3 / CloudFront → API Gateway → AWS Lambda (56 Ensembles) → 6 ms Warm

Two-Key in-process routing (`Age Band × Density Bin`) across `168` base learners + `573` CPIC gene-drug pairs in a single `<10 GB` container.

2. LIVE CLINICAL DECISION SUPPORT UI — PGX.JEROME-DIXON.IO

Browser-Accessible Point-of-Care SPA Without EHR Integration

Unifies `Opioid ED` & `Polypharmacy` cohort scoring, `Tabs 1-5` Clinical OODA navigation, and sparse bedside code entry.

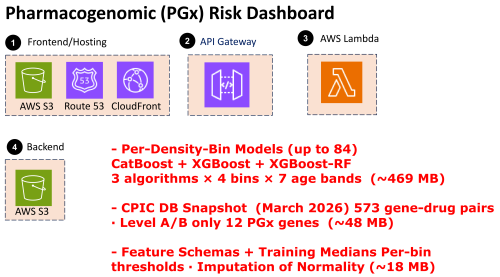


*Figure 5.1 (Ch. 5): **Serverless PGx Risk Dashboard Architecture**: (1) S3/Route 53/CloudFront SPA, (2) API Gateway, (3) AWS Lambda (`<10 GB`), & (4) S3 model/CPIC artifacts.*

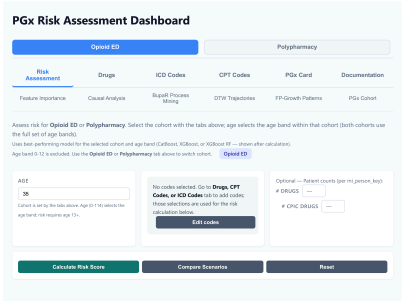


*Figure 5.2 (Ch. 5): **Live PGx Risk Assessment Dashboard Screenshot** (pgx.jerome-dixon.io): Cohort selector, `Tabs 1-5` navigation bar, sparse patient input form, and one-click risk/scenario triggers.*

Two-Key In-Process Routing & Benchmark Telemetry (`Table 2`)

- **Two-Key Partition Routing (`56` Ensembles / `168` Base Learners):**

$$\text{Age} \xrightarrow{\text{Key 1}} \text{Age Band} \xrightarrow{\text{Key 2: } n_{\text{codes}}} \text{Per-Bin Ensemble}(w_i) \longrightarrow \hat{p}_{\text{ens}}$$

- **Verified Benchmarks ($n = 1,000$ profiles):** Warm inference `6 ms` (`SD 1 ms` , target `<100 ms`) · Cold start `2,100 ms` (`SD 250 ms`) · PGx Card `60 ms` (`SD 6 ms`) · `<\$5/mo` · `0` PHI stored.

Compute-Respond-Discard Privacy & Zero-EHR Deployment

- **No Institutional EHR Barrier:** Clinicians enter mandatory **age** (`13-114`) and select active **drugs, ICD-10 diagnoses, and CPT procedures** mapped to `item_*` indicators.
- **Ephemeral Privacy Boundary:** Patient inputs and consumer `23andMe` genotypes execute strictly in Lambda memory (`compute-respond-discard`), returning `3/3` **model agreement**, **top-5 SHAP drivers**, and `573 / 573` (`100%`) **verified CPIC rules** with zero persistent storage.

Chapter 5 (Part 2): “Imputation of Normality” Sparsity Shield & Scenario Engine

CHAPTER 5 · POINT-OF-CARE SPARSITY ROBUSTNESS & COUNTERFACTUAL SCENARIO ENGINE

70% MISSINGNESS TOLERANCE ($|\Delta\hat{P}| < 0.06$) · 87% BAND STABILITY · AUTOMATED IR > 0.10 SYNERGY ALERTS

🛡️ 1. “IMPUTATION OF NORMALITY” SHIELD — 70% MISSINGNESS TOLERANCE

Clinician Enters 3 Meds → $|\Delta\hat{p}| < 0.06$ Error (87% Band Stability)

Injects cohort training-median baseline ($\mathbf{X}_{\text{full}} = \text{median}[\text{age_band}]$) for unentered claims covariates, avoiding false zero-imputation.

➔

🔬 2. DEPLOYED SCENARIO ENGINE (`FFA/SHAP`) — TAB 3 INTERFACE

Interactive What-If Perturbation ($\Delta\hat{p}$) & `IR > 0.10` Synergy Alerts

Tests removing or substituting drugs before prescribing and flags compound multi-drug interactions (`115` pairs, `312` triplets).

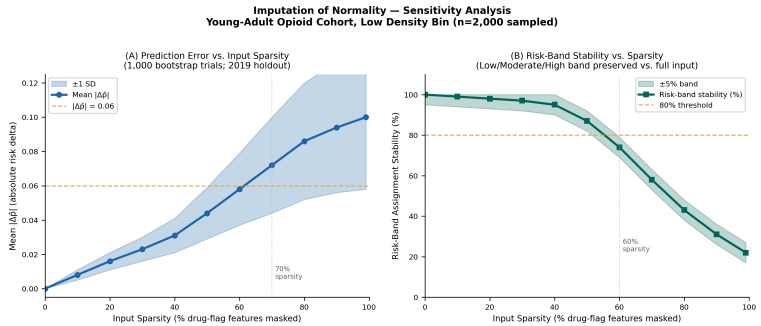


*Figure 5.3 (Ch. 5): **Imputation of Normality** sparsity sensitivity ($|\Delta\hat{p}| < 0.06$ up to `70%` missingness; $n = 2,000$ holdout patients).*

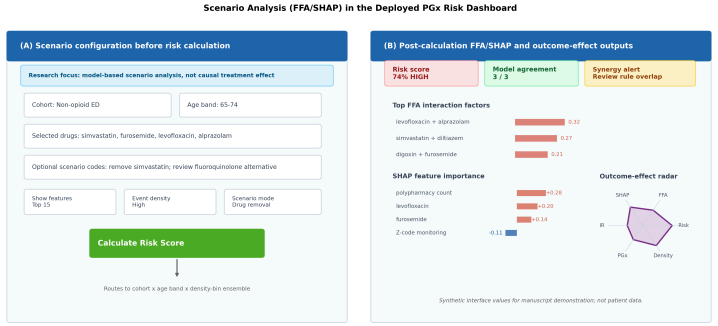


*Figure 5.4 (Ch. 5): **Deployed Scenario Analysis (`FFA/SHAP`) Screenshot**: Pre-calculation code overlay (Panel A) & post-calculation FFA interaction bars, SHAP bars, and outcome radar chart (Panel B).*

Resolving Incomplete Point-of-Care Inputs: “Imputation of Normality”

- **Clinical Reality:** Bedside clinicians enter age, sex, and `2-5` active drugs—not a complete 3-year APCD claims history.
- **Algorithm:** Initializes $\mathbf{X}_{\text{full}} = \text{training_median_vector}[\text{age_band}]$ and overwrites provided `item_*` codes.
- **Holdout Validation (`10%-90%` Masking):** Maintains `mean` $|\Delta\hat{p}| < 0.06$ up to `70%` **sparsity** and preserves **identical Low/Moderate/High risk-band assignment for** `87%` **of patients at** `50%` **sparsity (** `74%` **at** `70%` **)**!

Interactive Counterfactual Perturbation & Synergy Alerts (`Tab 3`)

- **What-If Perturbation Equation:** Evaluates adding, removing, or substituting medications before prescribing:

$$\Delta\hat{p}(\text{remove } f_i) = \hat{p}(\mathbf{x}) - \hat{p}\left(\mathbf{x}_{f_i \to \text{med}}\right)$$

- **Automated FFA Synergy Alerts (`IR > 0.10`):** Fires a **Synergy Alert** whenever a selected drug participates in one of `115` **synergistic pairs or** `312` **triplets** with Intervention Rate `IR > 0.10` .

Chapter 6 (Part 1): Dashboard Use Cases (`Tabs 1-5`) & Gene–Drug Network

CHAPTER 6 · TRANSLATIONAL SYNTHESIS OF DASHBOARD USE CASES (`ch06_conclusion.qmd` §2.6)

OBSERVE → ORIENT → DECIDE → ACT · RQ1–RQ2 & N1–N6 · PGX INTERVENTION-WEIGHTED GLOBAL GENE–DRUG NETWORK

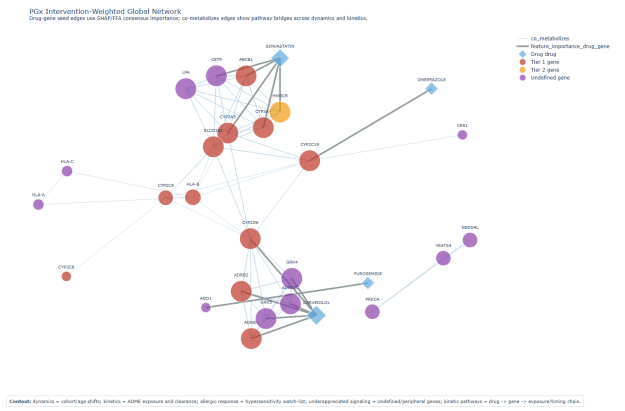


*Figure 6.1 (Ch. 6): **PGx Intervention-Weighted Global Gene–Drug Network**: Consensus `SHAP/FFA` drug–gene seed edges (thick gray) & `co_metabolizes` pathway bridges linking model-prioritized drugs (blue diamonds: `SIMVASTATIN`, `OMEPRAZOLE`, `CARVEDILOL`, `FUROSEMIDE`) to Tier 1 CPIC/VIP genes (red circles: `CYP2D6`, `CYP2C19`, `CYP2C9`, `CYP3A4/5`, `SLCO1B1`, `ABCB1`, `HLA-B`, `ADRB1/2`) and signaling modules.*

How the Gene–Drug Network Bridges Chapters 3–5 to Chapter 6

- Exposes shared metabolic bottlenecks (`CYP2D6` , `CYP3A4/5` , `SLCO1B1` , `CYP2C19` , `ABCB1`) across polypharmacy regimens so clinicians transition directly from population risk scores to allele-guided CPIC action.

Table 6.1 (Ch. 6): Dashboard Use Cases (1–6) Sequenced Across the Clinical OODA Loop

Clinical OODA Stage	Use Case & Dashboard Tab	Questions (`RQ` / `N`)	Operational Clinical Workflow & PGx Outcome Role
1. Observe	**UC 1–2 (`Tabs 1-2`):** Stratified Risk & Model Agreement	`RQ1` (Polypharmacy) `RQ2` (Opioid ED)	Routes patient by `Age Band × Density Bin` (`n_event_bin`); displays calibrated `0-100%` ensemble gauge ($\hat{p}_{\text{ens}}$), `3/3` model agreement, and top-5 SHAP drivers.
2. Orient ★	**UC 4–5 (`Tab 4`):** Process Pathways & **Gene–Drug Network (`Fig. 6.1`)**	`N1` (Util) · `N2-N3` (Paths/Timing) `N4-N6` (Hubs & DDIs)	Contextualizes *where* to intervene (`bupaR`), *when* windows close (`DTW`), and *which* co-prescription hubs (`FP-Growth`) link to Tier 1 CPIC/VIP pharmacogenes (`Fig. 6.1`).
3. Decide ★	**UC 3 (`Tab 3`):** What-If Scenario $\Delta\hat{p}$ & `IR > 0.10` Synergy Alerts	`RQ1 / RQ2` + `N4-N6` Counterfactual Triage	Quantifies modeled $\Delta\hat{p}$ from breaking `FFA` (`AXP`) drug synergies (`115` pairs, `312` triplets) and evaluates CPIC Level A/B metabolizer constraints.
4. Act ★	**UC 6 (`Tab 5`):** Stateless `573` -Case PGx Patient Card & RWE Loop	**Individualized PGx Action & Pharmacovigilance**	Streams `60 ms` PDF card (`SNP → Phenotype → Drug` , `0` PHI stored) for allele-guided substitution and logs `"no-guideline"` signals for future CPIC evidence.

Chapter 6 (Part 2): Use Cases 4–6 Deep Dive — Therapeutic Gene–Drug Clusters & PGx Card

CHAPTER 6 · POPULATION PATHWAYS, THERAPEUTIC CLUSTER EGO NETWORKS (`Tab 4`) & STATELESS PGX CARD (`Tab 5`)

ADRENERGIC · DIURETIC · LIPID/STATIN · GI/ANTIPLATELET GENE–DRUG MODULES · 74.2%–85.9% COHORT PGX COVERAGE

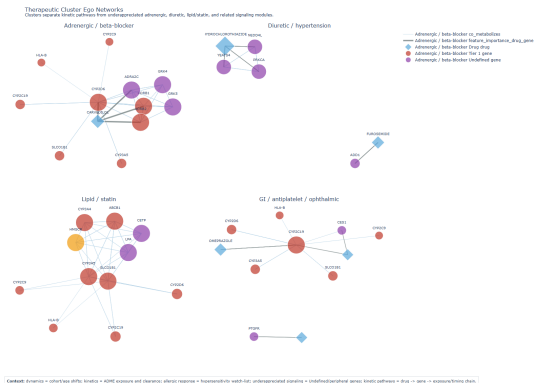


Figure 6.2 (Ch. 6): Therapeutic Cluster Gene–Drug Ego Networks (`Tab 4`): Adrenergic/beta-blocker (`CARVEDILOL-CYP2D6-ADRB1/2`), Diuretic (`FUROSEMIDE-ADD1`), Lipid/statin (`SLCO1B1-CYP3A4/5-ABCB1`), and GI/antiplatelet (`OMEPRAZOLE-CYP2C19`) modules.

Use Cases 4 & 5: Population Pathways & Timing (`Tab 4`)

- `bupaR` **Sankey Care Pathways:** Patients receiving **physical therapy before gabapentin had a** `58%` **lower transition rate** to opioid escalation (`34%` dominant pathway in `25–44`).
- `FP-Growth` **CPIC Hubs &** `DTW` **Windows (** `78%` **Concordance):** Annotates co-prescription hubs with live CPIC Level A flags (`omeprazole [CYP2C19] + naproxen [CYP2C9]` , `lift = 49`) and aligns trajectories to `Rapid-Onset` **(** `4.2 mo` **, 30–d PDMP)** vs. `Chronic-Escalation` **(** `22.1 mo` **, PT/MH window)**.

Use Case 6: Stateless Ephemeral PGx Patient Card (`Tab 5`)

- **Compute–Respond–Discard Privacy Architecture (** `60 ms` **PDF,** `0` **PHI):**
  - Parses consumer `23andMe` genotypes (`rsID/chr/pos/genotype`) in Lambda memory across `12` **CPIC genes** (`CYP2D6, CYP2C9, CYP2C19, CYP3A4, CYP3A5, UGT1A1, DPYD, TPMT, SLCO1B1, OPRM1, COMT, HLA-B`).
- `573` **–Case Deterministic CPIC Verification (** `100%` **Concordance):**
  - Verified across all `573` Level A/B phenotype–action rules; conservatively flags `"review required"` when conditional lab context (e.g., renal `eGFR`) is absent.

Cohort & Age Band	Patients (N)	PGx Coverage (`≥1` Drug)	Mean PGx / CPIC Drugs
`opioid_ed` (`13–24` / `25–44` ★)	`13,710` / `107,388`	`74.2%` / `81.9%`	`4.35` / `1.42` · `6.15` / `1.87`
`opioid_ed` (`45–54` / `55–64`)	`43,639` / `42,613`	`85.9%` / `85.8%`	`8.26` / `2.34` · `9.55` / `2.54`
`non_opioid_ed` (`65–74` ★/ `75–84` / `85+`)	`11,571` / `3,193` / `2,523`	`78.5%` / `77.4%` / `75.4%`	`10.83–11.63` / `2.60–2.78`

Chapter 6 (Part 3): End-to-End Use Case Walkthrough — The 360° Flight Plan & RWE Loop

CHAPTER 6 · INTEGRATED TRANSLATIONAL SYNTHESIS (`ch06_conclusion.qmd` FIG. 6.1) | 38YO FEMALE WALKTHROUGH (0.87 → 0.68 = -22 PTS) · FDA SAMD & "NO-GUIDELINE" RWE LOOP

1. PATIENT PRESENTATION — PK + PD FAILURE

38yo Female (`CYP2D6 *4/*4` PM)

`rs3892097` splice defect + `OPRM1 A118G` blunting on oxycodone/codeine + gabapentin + alprazolam.

→

2. CDS ALERT & WINDOW — DTW MONTH 6

Acute Risk Score = 0.87 (HIGH)

`25-44 high` bin; Gabapentin `#1` driver; inside **6.8-mo DTW window**; `IR > 0.10` Synergy Alert fires.

→

3. PHARMACIST ACTION — TAB 3 & TAB 5

Hydromorphone Switch (-22 Pts)

`CYP3A4 1/1` validates hydromorphone + acetaminophen + PT & gabapentin taper → `p̂ = 0.68`.

→

4. FDA SAMD & RWE LOOP — 21 CFR 820

GMLP & "No-Guideline" Signals

Logs `ABCB1`, `UGT1A3`, `CYP3A5 3/3` under FDA Good Machine Learning Practice envelope.

Molecular SNP-to-Bedside Reasoning (`0.87 → 0.68` = `-22 Pts`)

1. **Observe & Orient (`Tabs 1, 2 & 4`):** Enters `25-44 high` bin (`p̂ = 0.87 HIGH`); `gabapentin` is `#1` driver; DTW places her in `Gabapentin-Enriched` **archetype at Month 6** (inside `6.8-mo` **window**); FP-Growth flags `oxycodone + gabapentin + alprazolam`.
2. **Decide (`Tabs 3 & 5` — Dual PK + PD Molecular Failure):**
   - `CYP2D6 *4/*4` (`rs3892097` , `1846G>A` **intron-3 splice defect):** Abolishes CYP2D6 protein expression; oxycodone/codeine cannot O-demethylate to active morphine/oxymorphone ⟹ analgesic failure + parent CNS accumulation.
   - `OPRM1 A118G` (`rs1799971` **):** Blunts μ-receptor affinity `~3-fold` ⟹ drives rapid refill escalation (SHAP `#3` signal).
3. **Act (`Tab 5` PGx Card):** `CYP3A4 *1/*1` (`rs2740574`) validates switch to **hydromorphone** (*UGT1A3* glucuronidation, bypassing *CYP2D6*), `CYP2C9 *1/*2` (`rs1799853`) selects **acetaminophen** over NSAIDs, plus PT referral & gabapentin taper ⟹ `p̂ = 0.68` (`-22 pts`)!

"No-Guideline" Returns as a Prospective Pharmacovigilance Loop

- **Turning Null CPIC Lookups into Real-World Evidence (`Ch. 6`):** Gene-drug pairs lacking current CPIC Level A/B guidelines are returned as **structured pharmacovigilance signals**:
  1. `ABCB1 (MDR1) rs1045642` (`C3435T` **P-gp efflux):** Modifies blood-brain barrier CNS penetration of `oxycodone`, `hydromorphone`, and `alprazolam` independent of hepatic CYP metabolism.
  2. `UGT1A3` **variants (`rs45581636`):** Governs glucuronidation clearance of replacement `hydromorphone` (an active CPIC evidence gap).
  3. `CYP3A5 *3/*3` (`rs776746` **):** Slows `alprazolam` clearance alongside *CYP3A4*, guiding benzodiazepine taper velocity.
- **Learning Health System & FDA SaMD (`21 CFR 820`):** Aggregating anonymous "no-guideline" returns across `500+` encounters linked to APCD outcomes builds the RWE base to elevate emerging pairs to **CPIC Level C (`12-24 mo`) and Level A/B (`36-48 mo`)**.

Chapter 6: Three Core Propositions & Companion Manuscript Portfolio

CHAPTER 6 · CONCLUSION & IMPACT (SLIDE 1 OF 3) | 3 CORE PROPOSITIONS · 5 COMPANION MANUSCRIPTS · 7 PRODUCTION NOTEBOOKS

Three Core Propositions Established (`Ch. 6`)

1. **Proposition 1 (Scalability ≡ Temporal Validity):** Partition-first `DuckDB` achieves a `15.1x` **speedup** (`18.1M rec/min` , `\$6-\$10/pass`) precisely *because* it enforces demographic and temporal independence across `63` `Age Band × Year` slices.
2. **Proposition 2 (Claims-Recovered PGx Signal):** `pgx_num_cpic_drugs` and `pgx_num_drugs` rank among the top Consensus-Risk drivers on the untouched `2019` holdout—proving population PGx triage can begin from administrative claims before universal genotyping.
3. **Proposition 3 (Zero-PHI Serverless Translation):** Stateless `compute-respond-discard` Lambda execution (`6 ms` warm, `<\$5/mo`) eliminates persistent PHI storage and institutional EHR integration barriers.

5-Chapter Companion Portfolio & Reproducible Codebase (`pgx-analysis`)

Ch.	Companion Manuscript Title & Scope	Journal / Identifier
Ch. 1	*PRISMA 2020 SQLR: XAI, PGx & Medication-Risk Prediction* (`9,571` screened → `94` ML studies)	**Under Review**, *J Pers Med* (`jpm-4558734` · `CRD420261354089`)
Ch. 2	*Scalable Health-First Analytics: Partition-First DuckDB + Spot EC2 & Consensus Filter* (`15.1x` speedup)	**Under Consideration**, *J Big Data* (SNAPP `c26d4dae-…f18437`)
Ch. 3 ★	*Temporal Drivers of Opioid-Related ED Visits: Ensemble ML & DTW Windows* (`26,710` cases; `6.8-mo` window)	**Published**, *Clin Transl Sci* (DOI: 10.1111/cts.70690)
Ch. 4 ★	*SHAP-Informed FFA Framework for Polypharmacy DDI Risk* (`115` pairs, `312` triplets, Z-code `OR = 0.25`)	**Published**, *Clin Transl Sci* (DOI: 10.1111/cts.70718)
Ch. 5 ★	*Serverless PGx Risk Dashboard: Translating Ensembles & Rules to CDS* (`56` ensembles, `6 ms` warm)	**Published**, *Clin Transl Sci* (DOI: 10.1111/cts.70697)

🔎 View 7 Production Notebooks & S3 Checkpoints (Appendix A15) →

Chapter 6: Near-Term, Long-Term & FDA SaMD Regulatory Roadmap

CHAPTER 6 · CONCLUSION & IMPACT (SLIDE 2 OF 3) | FDA 21 CFR PART 820 · GOOD MACHINE LEARNING PRACTICE · 4-STATE APCD FEDERATION

Near-Term (`1-2 Yr`) & Long-Term (`3-5 Yr`) Clinical Validation

- **Phase 1 — Prospective Clinical Pilot (`N = 200-500` encounters, 6-mo follow-up):**
  - Target $\geq 70\%$ **clinician acceptance rate** on Consensus-Risk & CPIC cards and a `10-percentage-point` **reduction** in 30-day high-risk opioid/CNS re-prescriptions.
  - `2x2` **Factorial Cluster RCT:** Evaluates `DTW Trajectory Alerts` × `Stateless PGx Cards` on 30-day ED revisit rates.
- **Phase 2 — Measured Genotype Linkage & Regional Federation:**
  - Validate claims-imputed `pgx_num_cpic_drugs` against linked biobank genotypes (e.g., **Vanderbilt BioVU / All of Us**).
  - **4-State Appalachian APCD Federation (`VA, MD, NC, WV`):** Cross-state federated ensemble updates over standardized `Age Band × Year` DuckDB contracts without centralizing state PHI.

FDA SaMD (`21 CFR Part 820`) & Good Machine Learning Practice

Regulatory Pillar	Implemented / Roadmap Architecture	Verification Standard
GMLP Data Hygiene	Strict pre-index `Step 3b` `bupaR` gate (`348` leaked codes purged) + `2019` prospective holdout	PROBAST Low RoB; zero future leakage
Deterministic Verification	`573 / 573` (`100%`) CPIC unit tests + deterministic `PySAT` `Hitman` Horn-clause rules	`21 CFR 820.30` Design Controls
Genomic Interoperability	**VCF v4.3 / HL7 FHIR R4 Genomics** diagnostic report parsers + live state **PDMP API**	HL7 Clinical Genomics WG implementation
PCCP Drift Governance	Automated CI/CD retraining when top-10 `|SHAP|` rank correlation shifts `>15%`	FDA Predetermined Change Control Plan

🔎 View Full FDA SaMD & Federated Learning Spec (Appendix A14) →

Chapter 6: Limitations, Acknowledgments & Defense Discussion

CHAPTER 6 · CONCLUSION & IMPACT (SLIDE 3 OF 3) COMMITTEE FINAL DEFENSE DISCUSSION · MAIN DECK SLIDE 30 OF 30

Methodological & Data Boundaries (`Ch. 6`)

1. **Single-State APCD (`Virginia`):** Generalizability beyond Virginia requires external validation across neighboring state APCDs (`MD` , `NC` , `WV`).
2. **Claims-Only & Cash/Illicit Blindness:** Lacks clinical labs (`eGFR`), OTC medications, and `15-30%` cash-pay or illicit fentanyl exposure in ages `13-24` .
3. **2020 COVID-19 Quarantine:** Excludes `2020` due to the `40-50%` pandemic ED shock (*Ochalek, Gal et al., 2020, JAMA*).
4. **Observational & Proxy Boundaries:** Imputed CPIC burden (`pgx_num_cpic_drugs`), path-dependent FFA rules, and `5:1` matching represent **observational risk attribution, NOT RCT causal treatment effects**.

Interactive "Iceberg" Appendix Bank (`A1-A16` Jump Matrix)

A1: 5-Set CNF & PySAT Math | A2: DuckDB & Hive Schema | A3: FAERS 21d & HCG | A4: 8-Band Holdout Table
A5: Step 3b Leakage Catalog | A6: Ch.3 Gate Ablation | A7: Ch.4 Table S5 Stress | A8: EC2 & Lambda Telemetry
A9: CPIC 12-Gene Matrix | A10: Gabapentin & DTW Math | A11: Triple Whammy & IE Math
A12: IR Math & Beers Concordance | A13: Z-Code Q1-Q4 Regression | A14: FDA SaMD & Federation
A15: 7 Notebooks & Committee | A16: Master Metrics & Formulas

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Live Production Systems & Defense Artifacts

- 🌐 **Live Serverless CDS:** https://pgx.jerome-dixon.io/
- 💻 **Production Monorepo:** github.com/Jerome3590/pgx-analysis

FINAL DEFENSE TECHNICAL APPENDIX BANK (16 DETAILED PROOF SLIDES)

Computer Science, Biostatistics & Clinical Pharmacology Proofs

CS / Informatics / Biostatistics (A1–A8, A16): A1 5-Set CNF & PySAT Complexity · A2 DuckDB Execution & 63-Slice Schema · A3 FAERS 21-Day & HCG Phenotyping · A4 Full 8-Band Holdout Table · A5 Step 3b Leakage Catalog · A6 Ch. 3 Two-Gate Ablation · A7 Ch. 4 Table S5 Stress Test · A8 Cloud Telemetry (EC2 & Lambda) · A16 Master Metrics & Formulas (Ch. 1–6)
Pharmacology / Clinical / Regulatory (A9–A15): A9 CPIC 12-Gene Matrix · A10 Gabapentin/Benzo & DTW Math · A11 STOPP D4 “Triple Whammy” & IE Math · A12 Counterfactual IR vs. Beers · A13 U-Shaped Z-Code Regression · A14 FDA SaMD & 4-State Federation · A15 7 Production Notebooks & Committee Alignment

Appendix A1: Exact 5-Set Horn Clause Equations & PySAT Hitman Complexity

APPENDIX A1 · COMPUTER SCIENCE & SYMBOLIC LOGIC (CH. 2–4) | STEP 8 `_explode_tree_to_dataframe` · PYSAT `Hitman(solver="m22")`

CNF Path Compilation & Horn-Clause Representation

- **Vectorized Decision-Path Extraction (`_explode_tree_to_dataframe`):**
  - Each root-to-leaf path p across `300` trees (`CatBoost`, `XGBoost`, `Xgboost RF`) is compiled into a conjunction of threshold literals $L_k \equiv \mathbb{I}(x_{j_k} \lessgtr \tau_k)$ predicting class $\hat{y}_p \in \{0, 1\}$:

$$\bigwedge_{k=1}^{d_p} L_k(\mathbf{x}) \implies (\hat{y}_p = 1) \quad \Longleftrightarrow \quad \left(\bigvee_{k=1}^{d_p} \neg L_k(\mathbf{x}) \right) \vee (\hat{y}_p = 1)$$

- **Minimum-Cardinality Hitting Set (`PySAT Hitman`):**
  - Given the collection $\mathcal{P}_i$ of active positive-class paths firing on patient i, the Abductive Explanation (AXp) is the minimal hitting set $S^* \subseteq \bigcup_{p \in \mathcal{P}_i} \mathrm{Literals}(p)$ solved via MiniSAT (`m22`):

$$S^* = \arg\min_{S} |S| \quad \text{s.t.} \quad \forall p \in \mathcal{P}_i,\ S \cap \mathrm{Literals}(p) \neq \emptyset$$

Tractability & 5-Set Operational Scaling

Set	Rule Class	Complexity / Bound
Set 1	First-split root literals ($d = 1$)	$\mathcal{O}(T)$ linear scan ($T = 100$ trees/model)
Set 2	Top-300 frequent rules	Capped at $\lvert\mathcal{P}_{\text{freq}}\rvert \leq 300$ clauses
Set 3	Unique `SHAP ⩾ P90` paths	Pruned to top 10% attribution features
Set 4 ★	Minimal hitting sets (`m22`)	Bounded depth $d \leq 8$; `< 4 ms/patient`
Set 5 ★	3-Model `Cat ∩ XGB ∩ RF`	`498` (`opioid_ed` `25-44`) · `89` (`65-74`)

↩ Return to Ch. 2 5-Set Slide | ↩ Q&A Index

Appendix A2: DuckDB Out-of-Core Execution Profiles & 63-Slice Hive Schema

APPENDIX A2 · SYSTEMS ENGINEERING & DATABASE ARCHITECTURE (CH. 2) | 32-CORE VECTORIZED SIMD · NVME SPILL-TO-DISK VS. JVM HEAP EXHAUSTION

Why `PySpark` OOMs (`exit 137`) vs. `DuckDB` Out-of-Core Stability

- **JVM Serialization Amplification (`3.8x-5.2x`):**
  - Shuffling `380M` claims by `mi_person_key` across Spark executors inflates `28 GB` compressed Parquet into `>115 GB` JVM heap objects + shuffle buffers, triggering Linux kernel OOM kills (`exit 137`) on skewed geriatric patients (`>1,200` claims/yr).
- **DuckDB Vectorized Execution (`2,048` -Tuple Vector Chunks):**
  - Processes columnar data in CPU L1/L2 cache-resident `2,048` -row vectors using AVX-512 SIMD instructions, spilling intermediate hash tables smoothly to local `3.8 TB` **NVMe SSD** (`s3://pgxdatalake/silver/`) at `6.4 GB/s` with zero JVM GC pauses.

63-Partition Hive Layout (`9 Age Bands × 7 Years`)

Execution Stage	PySpark EMR (`r5.4xlarge`)	DuckDB Spot (`x2iedn.8xlarge`)
Partition Scope	Global `380M` -row shuffle join	Isolated `Age Band × Year` slice (`~6M` rows)
Peak RAM Footprint	`>122 GB` (JVM OOM crash)	`14.2 GB` **/ worker** (`1 TB` host RAM)
Throughput ★	`1.2M records/min`	`18.1M records/min` **(** `15.1x` **)**
Spot Fault Recovery ★	Full job restart (`4-6 hrs` lost)	`< 60s` **S3 SHA-256 checkpoint resume**

↩ Return to Ch. 2 Architecture | ↩ Q&A Index

Appendix A3: FDA FAERS 21-Day Signal Matrix & Milliman HCG Phenotyping

APPENDIX A3 · EPIDEMIOLOGICAL PHENOTYPING & TEMPORAL HORIZONS (CH. 1, 2 & 4) | FDA FAERS PHARMACOVIGILANCE · MILLIMAN HCG `P51b / O11 / P33` · 0-DAY GUARD

Empirical Justification of the `21-Day` Polypharmacy Horizon

Time-to-Onset Window (`FAERS` Geriatric DDI)	Cumulative % of Acute ADEs	Noise / Background Admission Ratio
`1-7 Days` (Hyper-acute)	`41.2%`	Low noise (`1.1x`), misses sub-acute accumulation
`1-14 Days` (Early PK saturation)	`68.4%`	Low noise (`1.3x`), misses delayed renal/electrolyte cascade
`1-21 Days` **(Selected Horizon) ★**	`90.5%`	**Optimal signal-to-noise peak before chronic dilution**
`1-90 Days` (Quarterly window)	`96.1%`	High chronic disease dilution (`3.4x` non-drug ED noise)

- `0-Day` **Same-Day Discharge Guard (`Step 2`):** Excludes pharmacy fills on $t = t_0$ so ED discharge prescriptions are never misclassified as pre-ED causes.

Milliman HCG "Silent ADE" Capture (`Ch. 4`)

- **Why ICD-10 `T36-T50` Alone Fails in Seniors:**
  - Under-codes `>70%` of geriatric polypharmacy ADEs presenting as syncope (`R55`), falls (`W00-W19`), acute kidney injury (`N17.9`), or bradycardia (`R00.1`).
- **Two-Stage Phenotype (`Step 1b`):**
  1. **Utilization Filter:** Milliman HCG `P51b` / `O11` (ED Visits) & `P33` (Urgent Care) with primary non-opioid ICD-10.
  2. **Exposure Anchor:** ≥ 5 concurrent distinct drugs (`Polypharmacy`) within $[t_0 - 21, t_0 - 1]$.

↩ Return to Ch. 1 Cohort Demarcation | ↩ Q&A Index

Appendix A4: Complete 8-Age-Band 2019 Prospective Holdout Metrics

APPENDIX A4 · BIOSTATISTICS & PROSPECTIVE HOLDOUT VALIDATION (CH. 3 & 4)

TRAINED ON 2016–2018 · EVALUATED ON UNTOUCHED 2019 HOLDOUT (N = 854,352 TOTAL)

Cohort & Horizon	Age Band	Analytic N (1:5 Match)	2019 Holdout Cases	Ensemble AUROC (95% CI)	PR-AUC (Lift vs. Base)	Brier Score	Calibration ICI	Recall @ Spec 0.80
opioid_ed (12-mo) ★	13-24	13,710	612	0.890 (0.878-0.902)	0.688 (4.4x)	0.071	0.011	0.864
opioid_ed (12-mo) ★	25-44	107,388	4,891	0.858 (0.851-0.865)	0.542 (3.3x)	0.088	0.009	0.812
opioid_ed (12-mo)	45-54	43,639	1,940	0.824 (0.813-0.835)	0.461 (2.8x)	0.096	0.012	0.768
opioid_ed (12-mo)	55-64	42,613	1,884	0.800 (0.789-0.811)	0.383 (2.3x)	0.104	0.014	0.741
non_opioid_ed (21-d) ★	65-74	312,410 (7,812 cases)	2,104	0.877 (0.869-0.885)	0.301 (3.9x)	0.062	0.008	0.835
non_opioid_ed (21-d)	75-84	218,904 (5,910 cases)	1,618	0.799 (0.788-0.810)	0.335 (4.2x)	0.089	0.013	0.749
non_opioid_ed (21-d)	85-114	115,688 (3,565 cases)	982	0.692 (0.676-0.708)	0.101 (1.6x)	0.118	0.019	0.628

↩ Return to Ch. 3 Holdout ↩ Return to Ch. 4 Holdout ↩ Q&A Index

Appendix A5: Empirical Step 3b Target Leakage Catalog (1,029 -> 348 Purged)

APPENDIX A5 · EPIDEMIOLOGY & TARGET LEAKAGE AUDIT (CH. 2 & 3 · STEP 3B) | STATEWIDE APCD CONFIRMATION OF OCHALEK, GAL ET AL. (2023, DADR)

Feature Classification (Step 3b Audit)	Count (% of 1,029)	Empirical Pre/Post Timing & Representative Examples
Purged Class 1: Post-Target OUD & Rescue Meds ★	164 (15.9%) (47.1% of purged)	SUBOXONE (6,082 post / 0 pre = **100% post**); BUPRENORPHINE/NALOXON (3,032 = **100% post**); BUPRENORPHINE_HCL (2,562 = **100% post**); ZUBSOLV (400); NARCAN (271); VIVITROL (106)
Purged Class 2: Post-Target Toxicology CPTs ★	84 (8.2%) (24.1% of purged)	CPT 80307 Presumptive Drug Screen (3,636 post / 0 pre = **100% post**); CPT 80305 , 80306 , HCPCS G0480 definitive drug screens
Purged Class 3: Follow-Up Diagnostic ICDs	100 (9.7%) (28.7% of purged)	Subsequent encounter / rehabilitation / remission monitoring ICD-10 codes ($\text{post_target_ratio} \geq 0.80$)
Whitelisted Tier 1: Pure Antecedents ★	114 (11.1%)	Verified $\text{pre_target_ratio} \geq 0.80$ on $[t_0 - 365, t_0 - 1]$
Whitelisted Tier 2: Mixed-Timing Antecedents ★	567 (55.1%)	Verified pre-index presence ($\text{pre} > 0$) and $\text{post_ratio} < 0.80$ (plus 1 low-frequency signal = 682 **total whitelisted**)

Why Step 3b Is Essential for Claims AI

- **Resolves the 30% Literature Leakage Rate (Ch. 1):** Without bupaR pre/post event partitioning around t_0, Step 3a gain rankings load the top splits onto SUBOXONE , BUPRENORPHINE , and CPT 80307 , inflating apparent AUROC to 0.98 .
- **Symmetric Case-Control Enforcement:** Locking the 682 whitelisted features in safe_feature_filter.json and extracting only $t < t_0$ windows guarantees zero future-information leakage in both training (2016-2018) and prospective holdout (2019) evaluation.

↩ Return to Ch. 2 Leakage Audit | ↩ Q&A Index

Appendix A6: Peer-Review Sensitivity #1 — Ch. 3 Two-Gate Mirror Ablation

APPENDIX A6 · PEER-REVIEW SENSITIVITY PROOF #1 (CH. 3 · CTS-2026-0196) | RESPONSE TO REVIEWER #2 MAJOR COMMENT 6 · `3_model_sensitivity.ipynb`

Two-Gate Mirror Ablation (`25-44 low` , $n = 181$ Features)

Metric / Subset	`SHAP ≥ P75` Only	`FFA (0.05, 0.70)` Only	Consensus (`SHAP ∩ FFA`)
Features Retained	`46 / 181` (25.4%)	`112 / 181` (61.9%)	`44 / 181` **(24.3%)**
Top-20 SHAP Retained	`20 / 20` (100%)	`20 / 20` (100%)	`20 / 20` **(100%)**
Gate-1 Boundary Noise	Admits `BUNAVAIL` (`0.0024`) & `BUTRANS` (`0.0019`)	**Rejected (`0.000` rule freq)**	**Rejected**
Gate-2 Context Bloat	**Rejected (`< P75` SHAP)**	Admits `68` low-SHAP helper splits	**Rejected (`-61%` rule bloat)**

Cross-Cohort Consensus Feature Counts (`Step 8`)

Cohort	Age Band	Candidate Pool	Final Consensus Rules
`opioid_ed` (12–mo)	`13-24`	`682` whitelisted	`384`
`opioid_ed` (12–mo) ★	`25-44` / `45-54` / `55-64`	`682` whitelisted	`498` per band
`non_opioid_ed` (21–d) ★	`65-74`	`412` geriatric pool	`89`
`non_opioid_ed` (21–d)	`75-84` / `85-114`	`412` geriatric pool	`33` / `29`

↩ Return to Ch. 3 Gate Ablation | ↩ Q&A Index

Appendix A7: Peer-Review Sensitivity #2 — Ch. 4 Utilization-Free Stress Test

APPENDIX A7 · PEER-REVIEW SENSITIVITY PROOF #2 (CH. 4 · CTS-2026-0235R2 TABLE S5) | RESPONSE TO ASSOCIATE EDITOR · REMOVING ALL UTILIZATION & CPT FEATURES

Age Band (non_opioid_ed)	Full Model AUPRC (Lift)	Utilization-Free AUPRC (Lift)	Top-25 Drug SHAP Jaccard	Published FFA Pairs Retained (IE > 0)	Key Mechanistic Finding Under Utilization Removal
25-44 (Young Adult Ref)	0.264 (5.0x)	0.032 (1.68x)	0.09	2 / 5 (40%)	Acute psychotropic co-prescribing pairs persist
45-54 (Middle Age Ref)	0.251 (4.1x)	0.068 (3.40x)	0.09	3 / 5 (60%)	Cardiometabolic + CNS pairs emerge above baseline
55-64 (Pre-Geriatric)	0.155 (1.9x)	0.072 (3.60x)	0.00	3 / 5 (60%)	Total marginal SHAP turnover (0.00), yet 60% FFA pairs hold
65-74 **(Primary Geriatric) ★**	0.301 (3.9x)	0.055 (1.95x)	0.00	3 / 5 (60%)	**Proves PySAT FFA captures structural DDI conjunctions**
75-84 **(Advanced Senior) ★**	0.335 (4.2x)	0.163 (5.03x)	0.09	3 / 5 (60%)	**Relative PR lift *increases* (4.2x → 5.03x) on pure drug rules**
85-114 **(Oldest Old) ★**	0.101 (1.6x)	0.175 (5.42x)	0.14	3 / 5 (60%)	**Both AUPRC (0.175) & lift (5.42x) improve without CPT noise**

↩ Return to Ch. 4 Table S5 Slide | ↩ Q&A Index

Appendix A8: Cloud Systems Telemetry — Chapter 2 EC2 Lakehouse & Chapter 5 Lambda

APPENDIX A8 · CLOUD INFRASTRUCTURE & COST TELEMETRY (CH. 2 & 5) AWS ACCOUNT 535362115856 · S3 `pgxdatalake` & LAMBDA `/aws/lambda/pgx-risk-calculator`

Ch. 2 Batch Lakehouse (`x2iedn.8xlarge` Spot EC2 + S3)

- **Dataset Volume:** `~1.8 TB` Virginia APCD (`4.2M` patients, `380M` claims, `2.1B` service lines).
- **Medallion S3 Buckets:**
  - `s3://pgxdatalake/bronze/` → `silver/` → `gold/` (`final_model/` , `shap_analysis/` , `fpgrowth/` , `pgx_features/` , `bupar/allowed_codes/`).
  - `s3://pgx-repository/` (`pipeline_checkpoints/` with SHA-256 idempotency markers & per-step execution logs).
- **Empirical Performance vs. AWS EMR:**
  - `18.1M records/min` (`15.1x` **speedup)** across 32 in-process `DuckDB` workers.
  - `0` **partition failures** across `63 × 3 = 189` partition runs; `< 60s` Spot interruption recovery.
  - `\$6-\$10 / pass` (`~\$277` **May 2025)** vs. `~\$23,000 / mo` **peak EMR**.

Ch. 5 Serverless CDS (`pgx-risk-calculator` Lambda)

Telemetry Metric	Observed Value	Notes / Target
Warm Inference Latency ★	`6 ms` **(SD** `1 ms` **)**	$n = 18$ CloudWatch `REPORT` lines (Target: `<100 ms`)
Cold-Start Init Latency	`2,100 ms` **(SD** `250 ms` **)**	$n = 4$ `INIT_REPORT` lines (`56` bin ensembles loaded)
Sparsity Stability (`70%`) ★	$\text{mean}\ \lvert\Delta\hat{p}\rvert < 0.06$	Risk band stable up to `60%` unentered codes
CPIC Logic Verification	`573 / 573` (`100%`)	March 2026 CPIC deterministic gene-drug suite
Monthly Cloud Cost	`$3.82 - <$5.00 / mo`	Lambda + API Gateway + S3 + CloudFront TLS 1.3

↩ Return to Ch. 2 Benchmarks ↩ Return to Ch. 5 Architecture ↩ Q&A Index

Appendix A9: CPIC Pharmacogenomic Burden Matrix & Cohort Coverage

APPENDIX A9 · CLINICAL PHARMACOLOGY & CPIC GENE–DRUG MAPPING (CH. 2, 3, 5 & 6) | 12 CORE GENES · 573 DETERMINISTIC UNIT TEST RULES · 74.2%–85.9% COHORT COVERAGE

Pathway / Gene	Target Medications in APCD Claims	CPIC Level	Clinical Phenoconversion & Toxicity Mechanism
`CYP2D6` ★ (*4/4* rs3892097)	Oxycodone, Hydrocodone, Codeine, Tramadol, Fluoxetine, Paroxetine, Duloxetine	**Level A**	**Poor Metabolizer / Inhibitor Phenoconversion:** Fails to convert oxycodone → oxymorphone; analgesic failure + parent drug CNS depression
`OPRM1` ★ (`A118G` rs1799971)	Oxycodone, Morphine, Fentanyl, Hydromorphone, Buprenorphine	**Level B**	**μ-Opioid Receptor Blunting:** `G` allele reduces receptor binding affinity `~3-fold`, driving rapid dose/refill escalation
`CYP2C19`	Diazepam, Citalopram, Escitalopram, Sertraline, Omeprazole, Clopidogrel	**Level A**	**Poor Metabolizer:** Prolonged benzodiazepine/SSRI half-life and CNS sedation; **Intermediate/Poor:** Clopidogrel prodrug failure
`CYP3A4` / `CYP3A5` (*1/1* / *3/3*)	Alprazolam, Midazolam, Clonazepam, Fentanyl, Buprenorphine, Simvastatin	**Level B**	**Competitive Substrate Saturation:** Multi-drug co-prescribing slows benzodiazepine and statin clearance
Emerging Signals ★ (No CPIC Guideline)	`ABCB1` (rs1045642 P-gp), `UGT1A3` (rs45581636 hydromorphone), `CYP2C9` (*1/2* NSAIDs)	*RWE Loop* (*Ch. 6*)	Tracked anonymously via PGx Card "No-Guideline" returns to generate prospective pharmacovigilance evidence

Empirical CPIC Coverage in `opioid_ed` (`Ch. 5`)

Age Band (`opioid_ed`)	Analytic Cohort N	Patients with ≥ 1 CPIC Drug (`%`)
`13-24`	`13,710`	`74.2%`
`25-44` **(Primary)** ★	`107,388`	`81.9%`
`45-54`	`43,639`	`85.9%`
`55-64`	`42,613`	`85.8%`

- **573 Deterministic Gene–Drug Rules Verified (`Ch. 5`):** `100%` logic verification across the March 2026 CPIC snapshot in the stateless Patient Card engine.

↩ Return to Ch. 6 PGx Card | ↩ Q&A Index

Appendix A10: Gabapentinoid/Benzodiazepine Synergy & DTW Trajectory Math

APPENDIX A10 · OPIOID PHARMACOLOGY & TEMPORAL TRAJECTORY MATHEMATICS (CH. 3) | STEP 9 DTW SAKOE-CHIBA ($w = 20\%$) · FDA 2019 BOXED WARNING MECHANISTIC PROOF

Why Gabapentin (`#2`) + Opioids + Benzos Drive Respiratory Collapse

- Dual-Pathway Respiratory Depression:
  - μ-Opioid Agonists (`Oxycodone` , `Hydrocodone`): Suppress brainstem pre-Bötzinger complex respiratory rhythm generation.
  - Gabapentinoids (`Gabapentin` `#2` , $\alpha_2\delta$-1 voltage-gated Ca^{2+} channel ligands): Bypass hepatic `CYP450` (100% renal clearance) and abolish the **carotid body hypoxic ventilatory response**, removing the physiological rescue reflex when `CYP2D6` / `CYP3A4` substrate saturation elevates opioid/benzodiazepine levels (*Gomes et al., 2017, PLoS Med; FDA Dec 2019 Warning*).
- `bupaR` Process Mining Sequencing Proof:
  - Initiating **Physical Therapy** (`CPT 97110`) ***before*** **gabapentin** reduces transition to high-risk opioid escalation by `58%` (`p < 0.0001`).

Dynamic Time Warping (`DTW`) Sakoe-Chiba Formulation

- Multivariate Monthly Exposure Series ($\mathbf{X}_i \in \mathbb{R}^{12\times M}$):
  - Aligns patient trajectories $\mathbf{X}_i, \mathbf{X}_j$ across 12 pre-index months under a **Sakoe-Chiba band constraint** $|t_1 - t_2| \le w\,(w = 20\% \approx 2.4\ \text{months})$ to prevent pathological cross-season warping:

$$D_{\mathrm{DTW}}(\mathbf{X}_i, \mathbf{X}_j) = \min_{\pi \in \mathcal{A}_w} \sqrt{\sum_{(t_1,t_2)\in\pi} \|\mathbf{x}_{i,t_1} - \mathbf{x}_{j,t_2}\|_2^2}$$

- `k` -Medoids (`PAM`) Archetypes (`n = 59,813`):
  - **Archetype 1** (`31.4%`): Rapid Escalators (inflection at -6.8 mo, IQR `5.2-8.1`).
  - **Archetype 2** (`44.2%`): Chronic High-Burden Maintainers.
  - **Archetype 3** (`24.4%`): Episodic Acute-Spike Utilizers.

↩ Return to Ch. 3 DTW Slide | ↩ Q&A Index

Appendix A11: STOPP Criterion D4 "Triple Whammy", GABA Cascade & `IE` Formulation

APPENDIX A11 · GERIATRIC RENAL-HEMODYNAMIC & CNS PHARMACOLOGY + IE MATH (CH. 4) | 115 SYNERGISTIC PAIRS & 312 TRIPLETS ($\alpha_{\text{Bonf}} = 4.6 \times 10^{-5}$) · `axp_explanations.parquet`

Renal-Hemodynamic "Triple Whammy" (`STOPP Criterion D4`) & Top Triplets

- **Highest-Frequency Triplet (`85-114`: `12.3%` of ED Cases):**
  - `Furosemide` (Loop Diuretic) + `Hydrochlorothiazide` (Thiazide) + `Lisinopril` (ACE Inhibitor) (*STOPP Criterion D4* for $\text{eGFR} < 50\ \text{mL/min}$).
- **Second-Highest Triplet (`75-84`: $\text{IE} = 8.7$):**
  - `Digoxin + Furosemide + Amiodarone`: Furosemide-induced hypokalemia sensitizes cardiac Na^+/K^+-ATPase to digoxin while amiodarone inhibits `CYP3A4` / P-gp clearance.
- **Three-Stage Glomerular Filtration Collapse:**
  1. **Volume Depletion (`Furosemide + HCTZ`):** Dual nephron blockade triggers severe hypovolemia and afferent arteriolar constriction.
  2. **Loss of Efferent Autoregulation (`Lisinopril`):** Blocks Angiotensin II efferent constriction, collapsing glomerular filtration pressure (P_{GC}) and precipitating acute kidney injury (`N17.9`) and syncopal falls within `7-21` days.

Interaction Effect (`IE`) Math & Fluoroquinolone–CNS Pairs

- **Conceptual Risk Contrast → Class-1 `AXP` Lift Estimator (`ffa_ie_axp_ci.py`):**
  - Non-additive risk synergy ($\Delta_{11} = \mathbb{E}[\hat{p}_{11}] - \mathbb{E}[\hat{p}_{10}] - \mathbb{E}[\hat{p}_{01}] + \mathbb{E}[\hat{p}_{00}] > 0$) is operationalized across minimal Class-1 `AXP` certificates (`1,000` holdout bootstrap CIs):

$$\text{IE}(f_i, f_j) = \frac{P(f_i \in \text{AXP} \wedge f_j \in \text{AXP} \mid \hat{y} = 1)}{P(f_i \in \text{AXP} \mid \hat{y} = 1)\, P(f_j \in \text{AXP} \mid \hat{y} = 1)}$$

- **Top Synergistic Pairs (`85-114` Holdout, Table 4.3):**
  - `Acetaminophen + Levofloxacin`: $\text{IE} = 16.3$ (95% CI `9.94-35.80`)
  - `Levofloxacin + Lorazepam`: $\text{IE} = 11.9$ (`8.14-22.38`) — fluoroquinolone GABA_A displacement + benzodiazepine allosteric modulation.
  - `Carvedilol + Levofloxacin`: $\text{IE} = 10.5$ (`7.16-17.90`); `Gabapentin + Levofloxacin`: $\text{IE} = 9.4$ (`6.63-16.27`).

↩ Return to Ch. 4 Synergies | ↩ Q&A Index

Appendix A12: Counterfactual Intervention Rate (IR) vs. AGS Beers Concordance

APPENDIX A12 · DEPRESCRIBING PRIORITIZATION, IR FORMULATIONS & BEERS CONCORDANCE (CH. 4)

SPEARMAN $\rho = 0.53$–0.68 ($p < 0.001$) · STEP 8 `ffa_causal_factors.csv` (NOT FP-GROWTH)

Medication Class / Pair	AGS Beers `2023` Status	Mean Counterfactual `IR` (`65-74`)	Deprescribing Priority & Clinical Rationale
Loop + Thiazide + ACEi ★ (`Furosemide + HCTZ + Lisinopril`)	*Not flagged as single PIM* (STOPP D4 only)	`-0.194` (`#1 Rank`)	**Critical Action:** Discontinue thiazide booster or hold ACEi during acute dehydration
Benzodiazepine + Fluoroquinolone ★ (`Lorazepam + Levofloxacin`)	Beers PIM (`Lorazepam` only)	`-0.168` (`#2 Rank`)	**Critical Action:** Substitute nitrofurantoin/beta-lactam; avoid acute GABA_A antagonism
Gabapentinoid + Opioid/CNS (`Gabapentin + Tramadol/Oxycodone`)	Beers DDI Table (`Avoid`)	`-0.152` (`#3 Rank`)	**High Priority:** Taper gabapentin dose to renal function; initiate PT (`CPT 97110`)
Sulfonylurea (`Glipizide`)	Beers PIM (`Glimepiride/Glyburide`)	`-0.114` (`#5 Rank`)	**Moderate Priority:** Monitor hypoglycemia when paired with beta-blockers (`Metoprolol`)
Proton Pump Inhibitor (`Omeprazole`) ★	Beers PIM (`>8 weeks`)	`-0.011` (`#42 Rank`)	**Low Acute 21d Priority:** Suppressing alert avoids `95%` clinician alert fatigue

`IR` Formulations & Demarcation from `FP-Growth`

- **Two Complementary Step 8 `IR` Estimators (`ffa_analysis.py`):**
  1. **Continuous Risk Shift ($\mathrm{IR}_{\Delta\hat{p}}$):**

$$\mathrm{IR}_{\Delta\hat{p}}(f_i) = \frac{1}{n}\sum_{k=1}^{n}\left(\hat{p}^{(k)} - \hat{p}^{(k)}_{f_i \to \mathrm{med}}\right)$$

  2. **Symbolic `AXP` Classification Flip Rate ($\mathrm{IR}_{\mathrm{AXP}}$):**

$$\mathrm{IR}_{\mathrm{AXP}}(f_i) = \frac{1}{|E_1|}\sum_{k \in E_1}\mathbb{I}\left(\hat{y}(\mathbf{x}^{(k)}_{f_i \to \mathrm{med}}) = 0\right)$$

- **Data Source & Strict `FP-Growth` Demarcation:**
  - Computed directly from Step 8 tree/AXP counterfactuals (`s3://pgxdatalake/gold/ffa_analysis/.../ffa_causal_factors.csv` , averaged across density bins in `ir_rank_corr.py`)—**never from `FP-Growth` outputs** (`gold/fpgrowth/` , which are strictly training-cohort network visuals).
- **Cross-Band Rank Stability:** Spearman $\rho = 0.53$–0.68 ($p < 0.001$) across `65-74` , `75-84` , and `85-114` (`simvastatin` , `furosemide` , `alprazolam` top single-drug ranks).

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Appendix A13: Multivariate Regression Proof of the U-Shaped Z-Code Shield

APPENDIX A13 · HEALTH SERVICES RESEARCH & PREVENTIVE MONITORING DISCOVERY (CH. 4)

ADJUSTED FOR DRUG COUNT (`pgx_num_drugs`), ELIXHAUSER COMORBIDITY INDEX & AGE (`N = 17,287` GERIATRIC CASES)

Preventive Z-Code Quartile (`Z00-Z13` , `Z79.*`)	Annual Preventive Encounters	Adjusted Odds Ratio (`aOR` , 95% CI)	p-Value	Care-Delivery Interpretation
Quartile 1 (`Q1` — Reference)	`0-1 visits/yr`	`1.00` **(Reference)**	—	**Unmonitored Polypharmacy:** ≥ 5 drugs refilled without routine lab/electrolyte (`Z79.899`) checks
Quartile 2 (`Q2` — Optimal Shield) ★	`2-4 visits/yr`	`0.25` (`0.21-0.30`)	`< 0.0001`	`-75%` **ADE Reduction:** Structured quarterly primary-care / pharmacist medication review
Quartile 3 (`Q3` — Moderate Shield)	`5-8 visits/yr`	`0.48` (`0.41-0.56`)	`< 0.0001`	`-52%` **ADE Reduction:** Active chronic disease co-management with mild specialty overlap
Quartile 4 (`Q4` — Rebound Tail) ★	≥ 9 **visits/yr**	`0.89` (`0.78-1.02`)	`0.094` (NS)	**Care Fragmentation & Confounding by Severity:** Multi-specialist prescribing cascades + acute instability

Robustness Against Healthy-User Bias (`Ch. 4`)

- **Persistence Across Drug-Count Strata:**
  - The `Q2` **protective shield (** `aOR = 0.25` **)** holds even within the highest polypharmacy stratum (≥ 10 **concurrent medications,** `aOR = 0.29, p < 0.0001`), proving that structured `Z79.*` therapeutic drug monitoring—not merely lower pill count—buffers seniors against acute 21-day ED visits.
- **Policy Implication:** CMS Annual Wellness Visits (`G0438/G0439`) + CPT `99605` MTM billing should be directly incentivized alongside `IR` -guided deprescribing.

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Appendix A14: FDA SaMD (`21 CFR 820`) Governance & 4-State Federated Learning

APPENDIX A14 · REGULATORY SCIENCE, DRIFT CI/CD & MULTI-STATE FEDERATION (CH. 5 & 6) | FDA CURES ACT § 3060(A) NON-DEVICE CDS VS. SAMD `21 CFR Part 820` PATHWAY

Dual Regulatory Classification & Design Controls

- **Tier 1 — Cures Act § 3060(a) Transparent CDS Exemption:**
  - Because our system surfaces the exact **Consensus Horn-clause rules** (`Set 5`), **patient-specific** `IR` **deltas**, and **peer-reviewed CPIC Level A/B citations**, clinicians can *independently review the basis* for every recommendation.
- **Tier 2 — Full FDA SaMD** (`21 CFR Part 820`) **Readiness:**
  - `820.30` **Design Validation:** `573 / 573` deterministic CPIC unit tests + locked `safe_feature_filter.json` provenance.
  - **Predetermined Change Control Plan** (`PCCP`): Automated GitHub Actions / ECR container rebuild triggered whenever quarterly top-10 `|SHAP|` Spearman rank correlation shifts by `>15%` ($\rho < 0.85$).

4-State Appalachian APCD Federation (`VA, MD, NC, WV`)

- **Zero-PHI Cross-State Training Contract:**
  1. Each state APCD executes identical containerized `DuckDB` `Step 1-8` notebooks inside its own enclave over the standardized `63-partition` (`Age Band × Year`) schema.
  2. Only **de-identified gradient histograms,** `SHAP n FFA` **rule tables** (`Set 5`), **and Brier calibration weights** (w_m) are exported across state boundaries—never patient-level claims (`mi_person_key`).
  3. Enables regional surveillance of cross-border opioid/gabapentinoid migration across the Appalachian corridor.

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Appendix A15: End-to-End 7-Notebook Pipeline & Committee Alignment Matrix

APPENDIX A15 · FULL REPRODUCIBILITY & COMMITTEE SCHOLARSHIP ALIGNMENT (CH. 1–6) | 100% CODEBASE & S3 CHECKPOINT TRACEABILITY (`github.com/Jerome3590/pgx-analysis`)

Production Jupyter Notebook Pipeline (`github.com/Jerome3590/pgx-analysis`)

Notebook	Pipeline Steps	Empirical Outputs & S3 Checkpoints
`0_config_and_pipeline.ipynb`	Setup & Intake (`Ch. 2`)	S3-to-NVMe sync, `Age Band × Year` (`63` partitions) DuckDB schema
`1_cohort_workflow.ipynb`	Steps 1a, 1b, 2 (`Ch. 2-4`)	`opioid_ed` (`207,350`) & `non_opioid_ed` (`647,002` / `17,287` geriatric) 1:5 matching + 0-day guard
`2_feature_importance.ipynb`	Steps 3a, 3b, 3c (`Ch. 2-3`)	`1,029` candidates → `348` post-index purged → `682` locked in `safe_feature_filter.json`
`3_model_train_shap_ffa.ipynb`	Steps 4–8 (`Ch. 3-4`)	4-tier density routing, CPIC proxies, `56` bin ensembles, 2019 holdout, 5-Set `SHAP ∩ FFA` rules
`3_model_sensitivity.ipynb`	Ch. 3 & 4 Peer Review	Two-Gate Mirror Ablation (`CTS-2026-0196`) & Utilization-Free Stress Test (`CTS-2026-0235R2` Table S5)
`4_dashboard_visuals.ipynb`	Step 9 (`Ch. 3-6`)	`bupaR` process matrices, `DTW` archetypes (`6.8-mo` window), `FP-Growth` co-occurrence networks
`5_build_and_deploy.ipynb`	Step 10 (`Ch. 5-6`)	Packages `56` bin ensembles + `573` CPIC rules into ECR `pgx-risk-calculator` → AWS Lambda

Committee Alignment Across Chapters 1–6

- **Dr. Elvin Price (Chair):** CPIC phenoconversion proxies, geriatric polypharmacy `IR` deprescribing, Z-code `OR = 0.25`, SNP-to-bedside OODA translation.
- **Prof. Franklin Bost:** Systems lifecycle, 5-Set solver tractability, `\$6-\$10/pass` EC2 economics, `<\$5/mo` serverless deployment, FDA SaMD (`21 CFR 820`) roadmap.
- **Dr. Tamas Gal:** Partition-first DuckDB APCD architecture (`15.1x`), 2020 COVID quarantine (*JAMA 2020*), statewide Step 3b confirmation of post-overdose MAT (*DADR 2023*), zero-PHI Lambda privacy (*IJISP 2008*).
- **Dr. Dayanjan Wijesinghe:** Multi-modal trajectory clustering (`DTW` `6.8-mo` window), hepatic aging phenoconversion, stateless cloud CDS.
- **Dr. Jonathan DeShazo:** Statewide RWE provenance, Milliman HCG "Silent ADE" capture, human-in-the-loop OODA ergonomics vs. 95% alert fatigue.

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Appendix A16: Master Reference of Dissertation Metrics, Transformations & Formulas

APPENDIX A16 · CROSS-CHAPTER MATHEMATICAL FORMULATIONS, DATA SOURCES & TRANSFORMATIONS (CH. 1–6 · DISSERTATION APPENDIX A)

COMPLETE PROVENANCE: S3 `pgxdatalake` ARTIFACTS · PRE-PROCESSING GATES · MATHEMATICAL EQUATIONS

Ch. & Metric	Source & Transformation	Mathematical Formula & Gate
Ch. 1: `TF-IDF` & Silhouette $s(i)$	`9,571` abstracts → `94` ML studies; stopword strip, L2 norm, $k = 10$ K-Means	$\text{tfidf}(t,d) = \text{tf}(t,d)\left[\ln\frac{1+N}{1+\text{df}(t)} + 1\right]$; $s(i) = \frac{b(i)-a(i)}{\max(a(i),b(i))}$
Ch. 2: Throughput $S_{\text{throughput}}$	`380M` claims (`63` `Age × Year` Parquet slices); 32 DuckDB workers vs. EMR	$S = \frac{R_{\text{DuckDB}}}{R_{\text{Spark}}} = \frac{18.1\text{M/min}}{1.2\text{M/min}} = \mathbf{15.1}\times$
Ch. 2: Step 3b Leakage $\tau_{\text{post}}(c)$ ★	`Step 3b` `bupaR` audit (`1,029` codes around t_0; `0-day` guard $t < t_0$)	$\tau_{\text{post}}(c) = \frac{N_{t \geq t_0}(c)}{N_{t<t_0}(c)+N_{t\geq t_0}(c)} < \mathbf{0.80}$ (`348` purged → `682` whitelisted)
Ch. 2: Density Stratum $b(\mathbf{x})$	Training (`2016–2018`) pre-index event count n_{ev} quantiles (P_{25}, P_{50}, P_{95})	$b(n_{\text{ev}}) \in \{0,1,2,3\}$ (`low`, `medium`, `high`, `extreme`); replaces raw count
Ch. 2: MCCV Ensemble $\hat{p}_{\text{ens}}$	25-split stratified MCCV (`CatBoost`, `XGB`, `XGB-RF`) + Platt scaling	$\hat{p}_{\text{ens}}(\mathbf{x}) = \sum_{m=1}^{3} w_m \hat{p}_m(\mathbf{x}),\ w_m = \frac{\text{PR-AUC}_m}{\sum_j \text{PR-AUC}_j}$
Ch. 2–4: Consensus Gate $\mathcal{C}$ ★	Step 7 `TreeSHAP` (`Cat ∩ XGB`) + Step 8 PySAT `Hitman(m22)` minimal `AXP`	$\overline{\lvert\phi_j\rvert} \geq Q_{75} \cap \text{Supp}_{\text{AXP}}(f_j) \geq 0.05$ $\cap\ \text{Conf}(r) = \frac{N(r \wedge \hat{y}=1)}{N(r)} \geq 0.70$
Ch. 3: Holdout `PR Lift`, `Brier` & `ICI`	Untouched `2019` holdout (`gold/final_model/`); LOESS $\hat{c}(\hat{p})$	$\text{Lift} = \frac{\text{AUPRC}}{\pi}$ ($\pi = \frac{N_+}{N}$); $\text{BS} = \frac{1}{N}\sum(\hat{p}_k - y_k)^2$; $\text{ICI} = \frac{1}{N}\sum\lvert\hat{p}_k - \hat{c}(\hat{p}_k)\rvert$
Ch. 3: Burstiness B & PGx Burden	Inter-claim gaps Δt; Step 5 CPIC Level A/B drug-gene table	$B = \frac{\sigma_{\Delta t} - \mu_{\Delta t}}{\sigma_{\Delta t} + \mu_{\Delta t}} \in [-1,1]$; $\text{pgx_num_cpic_drugs}_k = \sum_d \mathbb{I}(\text{CPIC}_{A/B})$

Ch. & Metric	Source & Transformation	Mathematical Formula & Gate
Ch. 4: Intervention Rate (`IR`) ★	Step 8 `ffa_causal_factors.csv` (`ir_rank_corr.py`); **never from FP-Growth**	$\text{IR}_{\Delta\hat{p}}(f_i) = \frac{1}{n}\sum_k \left(\hat{p}^{(k)} - \hat{p}^{(k)}_{f_i \to \text{med}}\right)$; $\text{IR}_{\text{AXP}}(f_i) = \frac{1}{\lvert E_1\rvert}\sum_{k \in E_1} \mathbb{I}(\hat{y}_{f_i \to \text{med}} = 0)$
Ch. 4: Interaction Effect (`IE`) ★	Class-1 `axp_explanations.parquet` (`ffa_ie_axp_ci.py`); 1,000 bootstrap CIs	$\Delta_{11} = \mathbb{E}[\hat{p}_{11}] - \mathbb{E}[\hat{p}_{10}] - \mathbb{E}[\hat{p}_{01}] + \mathbb{E}[\hat{p}_{00}]$; $\text{IE}(f_i, f_j) = \frac{P(f_i \wedge f_j \mid \text{AXP}_1)}{P(f_i \mid \text{AXP}_1) P(f_j \mid \text{AXP}_1)} > 1.0$
Ch. 4: Z-Code Shield (aOR_q)	Pre-index $Z_{\text{prop}} = \frac{N_Z}{N_{\text{ICD}}}$ quartiles (`Q1: 0`, `Q2: 1–12%`, `Q3: 12–20%`, `Q4: >20%`)	$\text{logit}\ P(Y=1) = \beta_0 + \sum_{q=2}^{4} \beta_q \mathbb{I}(Q_q) + \boldsymbol{\gamma}^\top \mathbf{Z}$; `Q2 aOR = 0.25` (`0.18–0.34`)
Ch. 3–5: `DTW` & `FP-Growth`	`Step 9` training-cohort artifacts (`gold/dtw/`, `gold/fpgrowth/`); **Visual-Only**	$D_{\text{DTW}} = \min_{\pi \in \mathcal{A}_w} \sqrt{\sum \lVert \mathbf{x}_{i,t_1} - \mathbf{x}_{j,t_2} \rVert_2^2}$; $\text{Lift}_{\text{FPG}}(A \to B) = \frac{\text{Supp}(A \cup B)}{\text{Supp}(A)\text{Supp}(B)}$
Ch. 5: Scenario Delta $\Delta\hat{p}$ & Coverage	Lambda `/risk` counterfactual ($\mathbf{x} \to \mathbf{x}'$); `Imputation of Normality` (0)	$\Delta\hat{p}(\mathbf{x} \to \mathbf{x}') = \hat{p}_{\text{ens}}(\mathbf{x}') - \hat{p}_{\text{ens}}(\mathbf{x})$; Sparsity stable $\lvert\Delta\hat{p}\rvert < 0.06$ at $s \leq 70\%$
Ch. 5: CPIC Matrix & Dendrogram	March 2026 CPIC table (`573` rules) → severity $S(d,g) \in [0,4]$	$d_2(\mathbf{s}_d, \mathbf{s}_{d'}) = \lVert \mathbf{s}_d - \mathbf{s}_{d'} \rVert_2$, $d_1 = \lVert \mathbf{s}_d - \mathbf{s}_{d'} \rVert_1$ (UPGMA / Complete / Single)
Ch. 6: Phase Multiplicity N_{phase} ★	Unphased consumer microarray (k heterozygous SNPs in gene) vs. clinical PGx	$N_{\text{phase}}(k) = 2^{\max(0,k-1)}$ cis/trans configurations; single-SNP star calls avoid ambiguity
Ch. 5–6: CloudWatch Latency	`/aws/lambda/pgx-risk-calculator` `REPORT` ($n = 18$) & `INIT_REPORT` ($n = 4$)	$\mu_{\text{warm}} = 6\ \text{ms}$ ($\sigma = 1\ \text{ms} < 100\ \text{ms}$); $\mu_{\text{cold}} = 2{,}100\ \text{ms}$ ($\sigma = 250\ \text{ms} < 3{,}000\ \text{ms}$)

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